SPSS

GUIDE FOR BEGINNERS AND PROS

A Simplified Step-by-Step Guide to Data Analysis and Interpretation Using SPSS

Billy York

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Introduction

SPSS is an acronym that stands for Statistical Package for the Social Sciences. SPSS is prevalent in all works of life. For instance, it is used to determine the number of scholarly articles per year. The primary audience of SPSS is academic researchers.

Importance of SPSS

1. It is user-friendly, given that it was developed over 40 years ago.
2. It has a point and clicks interface.
3. You can share syntax with others.
4. You can use it to generate data from experiments.
5. You can save and reuse syntax.

Versions, Editions, And Modules

Versions are the release updates. There are 1-

28 versions now. Version 1 was developed in 1968 and was called SPSS. Version 24 came out in 2016. There is not much difference in each version because any code generated in one version can run in virtually any other version. Also, the version you are using is highly compatible with the file. Therefore codes generated in version 18 can be readable in version 26. This is because most of the features are consistent.

Editions talk about what is included in a purchase. They differ by price and function. There are four distinct editions. They are:

1. **Base Edition:** This edition gives you access to basic statistics, linear regression, clustering, and factor analysis.
2. **Standard Edition:** This gives you access to logistic regression, generalized LM, drag and drops active tables and Survival Analysis.
3. **Professional Edition:** This gives you access to data preparation, forecasting, decision tree, and

imputation.

4. **Premium Edition:** This gives you access to bootstrapping, complex sampling, exact tests, and SEM.

Modules

These are the components that add extra functionality to SPSS. The available modules are:

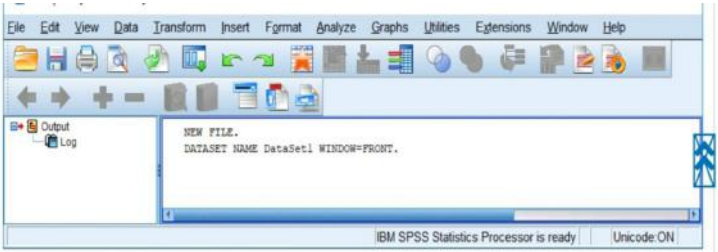
1. Advanced statistics.
2. Bootstrapping.
3. Categories.
4. Complex sampling.
5. Co joint.
6. Custom tables.
7. Data preparation.
8. Exact tests.
9. Forecasting.
10. Missing values.
11. Neutral networks.

Windows In SPSS

SPSS uses various types of windows in its basic operations. Each window is linked with specific tasks and types of SPSS files. The windows include:

1. **Output Viewer**

When you execute any command in SPSS, the Output Viewer window unlocks automatically and shows a log of the activities taken and the associated output. Mainly, the Output Viewer is where the outcomes of statistical analysis are expressed, but any command conjured through the drop-down menus or syntax will be printed in the Output Viewer. If an Output Viewer window is not open when order is run, a new Output Viewer window will automatically be generated.



Output Viewer window (SPSS Statistics version 24). The left panel outlines the commands and output already executed. The proper forum contains the tables and syntax.

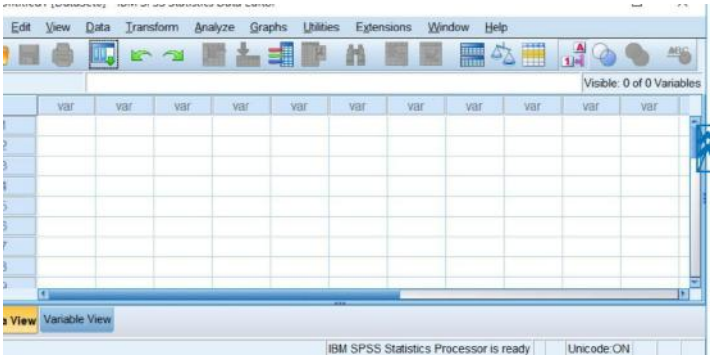
The Output Viewer window is split into two sections. The left section contains an overview of the content in the Output Viewer. This overview is useful when you have run several SPSS commands and want to locate a specific section of output quickly. The right section contains the actual production. Tapping on an item in the left branch will jump to that content in the right area. Things that have been chosen in the right team are signified by a red arrow and a box drawn around the content.

You can adjust the contents in the Output Viewer by choosing items in the left or right sections and pasting, copying, or deleting

them.

2. Data Rewriter

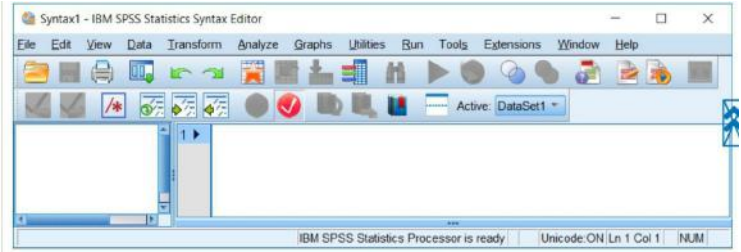
This window is usually the default window and opens automatically when SPSS starts. This window shows the content of any open data files and gives you access to the drop-down menus that allow you to adjust and analyze data. The data are offered in a spreadsheet format where columns stand for variables and rows stand for cases. The spreadsheet format comprises two tabs at the bottom tagged as Data View and Variable View. The Data View tab shows the open data set: variables are seen in the columns, and cases are seen in the rows. The Variable View tab exhibits data about factors in the accessible information like variable names, types, labels, etc. The color of the account that will be shown is yellow.



Data Rewriter window with Data View tab active (SPSS Statistics version 24).

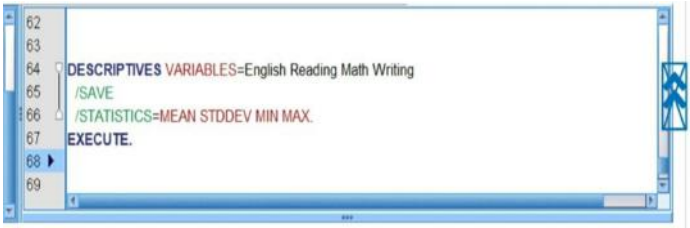
3. The Syntax Rewriter

SPSS syntax is a computer-aided learning language specific to SPSS, which can substitute the pull-down menus for data handling and sophistical analyses. The Syntax Rewriter window allows researchers to draft, execute, and debug SPSS syntax. To view a new Syntax Rewriter window, tap File > New > Syntax.



Syntax Rewriter window (SPSS Statistics version 24).

The right panel of the Syntax Rewriter window is where you put in your syntax. The left panel of the Syntax Rewriter window displays a feature of the commands in your syntax and can be used as a guide within your code. You can dive into a particular part of your code by tapping on the power in the left side panel. This element helps display the start and endpoints, mainly if the order is higher than one line.



The Syntax Rewriter window will show an overview of the navigation based on what

commands you use.

4. **Variable View -** The variable view comprises the variables on your data set. It is defined as the properties of your dataset. Each row will interpret all of the different variables for one data collection. Name, type, width, decimals, label, values, missing, columns, align and measure. Ensuring that the 'measure' of your variables is correct is essential. The variable can be Nominal for strings of data, Ordinal for data that is not consecutive but can be ranked, finally, scale, which is used for a straight variable, for instance, a distance to a place.

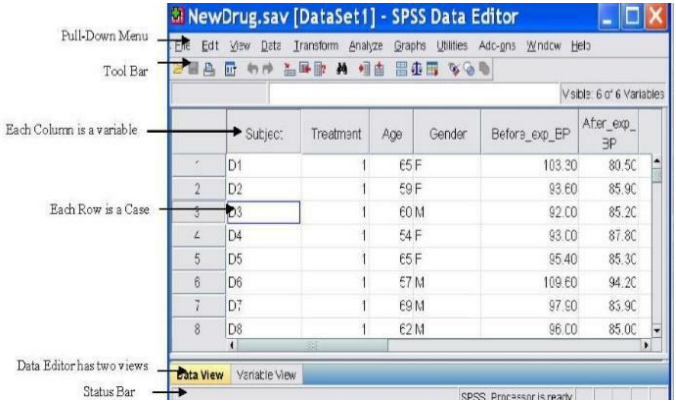
Active Windows

An active window is presently chosen and on which any commands will be run. This window is recognized by the tiny "+" figure

in the upper left corner of the window.

If you have various windows open, it is vital to know the active window because SPSS may instruct particular data to the active window. For instance, if there are one or two Output Viewer windows open and a procedure of command is in session, SPSS will publish the order outcomes in the present Output Viewer window. Therefore, if you want the output to show a different output file, ensure that file is the current viewing window.

Main Menus In SPSS



The Pull-Down Menus

Many tasks in SPSS are accomplished by selecting the right ``draw-down'' menus. Every window in SPSS has a unique menu bar with decent menu options and toolbars.

Graphs and Analyze menus are present in all windows. Below are Data Rewriter window menus and their functions.

1. **Data Menu:** This menu enables you to group cases, combine records, define variable properties, split files, select topics and utilize a variable to weigh cases.
2. **File Menu:** This feature opens existing records or a database record like an excel list or glancing at a text file. It also saves any modifications to the new file.
3. **Edit Menu:** This allows you to cut, paste, copy, insert cases, insert variables, or utilize find in the Data Rewriter window.

4. **Transform Menu:** This is where you see the options to perform some calculations on variables, form different variables from old ones.
5. **Analyze Menu:** All statistical analysis is done, from illustrative statistics to regression calculation to multivariate testing.
6. **Window:** The window menu enables you to alter the current window. The window that has a checkmark is the current one.

7. **Graphs Menu:** This feature lets you create high-resolution graphs and plots to be amended in the graph rewriter window or make interactive graphs.
8. **Utilities Menu:** This is used to show information on the scopes of SPSS data files or to handle the scripts.
9. **Add-Ons Menu:** From this menu, you can run other packages, for example, the conjoint succession

trees or Neural Networks. There are also program extensions that allow you to incorporate programs like R and Python into SPSS. If you wish to run any of the add-ons listed, you must purchase them.

10. **Help:** This is used to obtain assistance on issues in SPSS or get answers from the statistics coach to some fundamental problems.

Toolbars

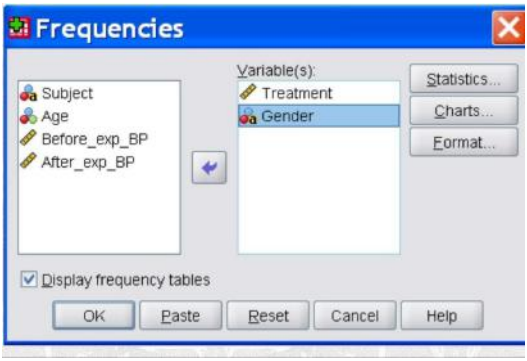
Every window in SPSS has its unique toolbars that access everyday tasks. When you lay the cursor on a tool, there is a brief explanation of what the device can do. You can move, show, or conceal a toolbar.

Status Bars

This is located underneath every SPSS window, and it gives the following information:

1. **Filter Status:** Filter On displays a fragment of cases in the data used for analysis.
2. **Command Status:** It provides information regarding a process currently running.
3. **Command Status:** It provides information regarding a process currently running.
4. **Weight Status:** Weight On shows that a weight variable is incorporated in the calculation.

Dialog Boxes



Several menu options open dialog boxes. Inside these dialogue boxes, you can select

variables and choices to be used. The parts of the main dialog box in any calculation process are:

1. **OK:** Tap this icon to kick-start the protocol.
2. **Paste:** Tap this button when you give out instruction syntax from your options. The instruction syntax is placed into a syntax window, from where it can be amended for future calculation. This generates the code known as SPSS projects.

3. **Source variable list:** This list of variable varieties (enabled by the process).
4. **Target variable lists:** These are numerous lists of variables required for the analysis.
5. **Command push buttons:** These are buttons employed in running the process. It is done by opening a sub dialogue box to create different

specifications. Below are some of the pushbuttons:

6. **Reset:** This undoes any options and restores all settings to their specifications in the dialog box and sub dialogue packages to the original status.
7. **Cancel:** This writes off any modification in the dialog box settings from when it was last opened. Clicking cancel will cause the dialog box to close.
8. **Help:** This assists with the current dialog box.

Icons And Their Functions

Icons	Tooltip	Description
	Find	Scan for a quan or remark in dataset or scan

		restore a value word in a set of d Initiate only whe section in the c view window chosen. **Edit> F and Edit>replac** ctrl+F and ctrl+ H
	Click on Variable	Go to a partic column (variabl the active dataset. **Edit>Go variable**
	Click on case	Go to a partic row (cases) in current dataset
	Undo a user activity	**Edit>undo** in pull-down menus tap ctrl+Z

	Redo a user activity	**Edit>redo or** Ct Y
	Recollect previously used dialogue	Displays the list most previously u dialogue windo Only recall w you want to re-ru similar analysis.
	Publish/ print	Print out the subj of the cur window. **File> Pr**
	Save a file article	Save the current c collection. **Fi Save or** Ctrl S
	Open the data article	Unlock a data f **File>Open> Dat**
	Variables	View all varia names, measurem

		levels, types, lab misplaced co and value tags in current window.
	Set variable	Put a new varia between occurring variab by default, r variables genera in this manner scale numeric. **Edit> Ins variable**
	Insert cases	Enter a case between two nota cases. **Edit> Ins cases**
	Run illustrative statistics	Run illustrative statistics on select variables using th

		frequency methoc The statistics demonstrated are deduced by the variable distributi status settings. Sc variables are outlined using me mode, range, standard deviatio median, minimun and maximum. Nominal and ordi variables are summarized with frequency table. Initiate only wher partition in the da view window is chosen.
	Select cases	Remove a set of r

		cases to a data based on sc measures or assig filter varia **Data> Select cas**
	Value labels	Ensure the raw c or the value tag shown in the c view wind **View> Valu Label**
	Split files	Classify y variables based an absolute varia for instance, if variable Gender chosen in the s file, runn illustrative statis on any ot variables

		generate descrip for males or fem singly.
	Utilize variable pairs	Assign or unassig pairs of variables display in the cur window. Numero pairs can be selec at any given time. **Utilities>Use variables Sets.** Note: You must f define variable teams. **Utilities> define variable s**
	Display all varieties	Display all variab in the current dataset. Initiate o if the dataset has

		been used. **Utiliti** **Display all variables**

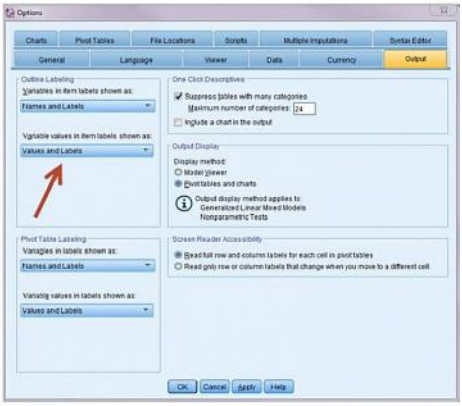
CHAPTER ONE

THE BASIC WORKSPACE

IBM SPSS statistics comes with a default setting of how the software works, how the data is handled and how the output is displayed.Most of the default settings may not be how you want them to be, you can also customize how SPSS works in order to adjust the look and feel of SPSS.

Customising SPSS Default Settings

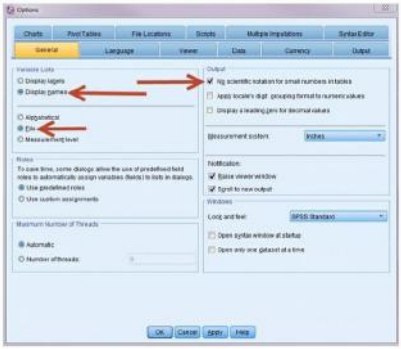
1. To Show the names and labels of variables in tables in the output. Go to Edit Options. Beneath the “Output Labels” tab, click on the pull-down menus to assign “Names”, “Labels”, or “Names and Labels” for each of the elements.



1. To have the variables shown in the way they appear in the data file rather than alphabetically in the lists of variables: Go to Edit Options. Below the “General” tab, in the upper left corner of the screen, tap on the radio lever to change from displaying the variables in alphabetical order to displaying the variables in the file order (see the screenshot below).
2. Displaying small numbers in decimals instead Of the scientific form in tables in the output: Go to Edit Options. Beneath the “General” tab, in the upper right corner, tap on

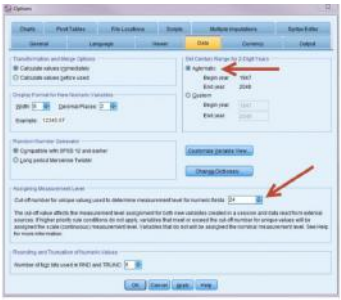
the box “No scientific notation for small numbers in tables' ' (see the screenshot below).

3. Setting SPSS to show the variable names instead of the variable labels in the list of variables: Go to Edit Options. Beneath the “General” tab, in the upper left corner, tap on the radio lever to change from displaying the variable labels to displaying the variable names.

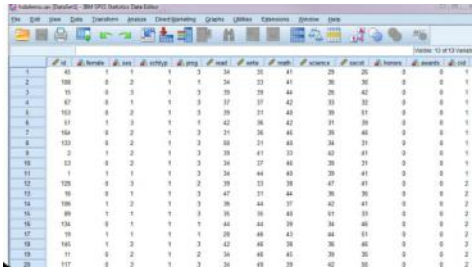


4. To change the number of distinct values used to specify the measurement status: Go to Edit Options. Below the “Data” tab, at the

bottom right corner, type in the minimum number of distinct values needed for a numeric variable for SPSS to analyze the variable scale. (see screenshot below).



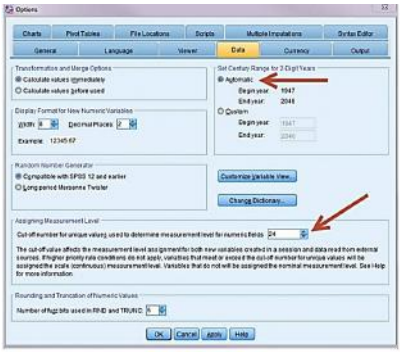
5. To get rid of the gridlines in the SPSS Data Editor: Go to View Uncheck “Grid Lines”.



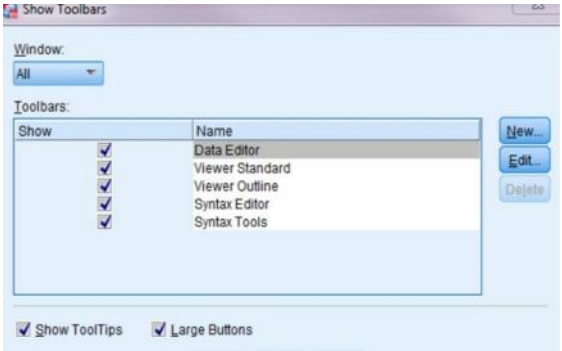
6. To display the syntax in the output

window: Go to Edit Options. Below the “Viewer” tab, in the lower-left corner, tap in the box to “Display instructions in the log.”

7. To alter the century cut-off date for two-digit years: Go to Edit Options. Beneath the “Data” tab, in the upper right corner, click on the radio lever “Custom” under “Set Century Range for 2-Digit Years”, and then type in the year that you need for the beginning year.

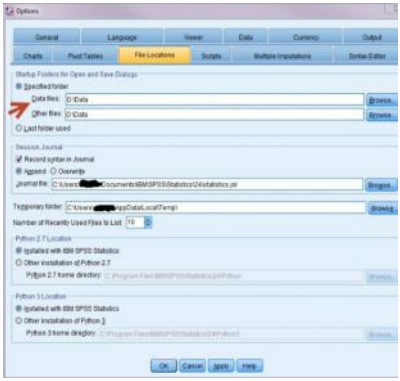


8. To customise the toolbars in the different SPSS windows: Go to View Toolbars. Then click on Customize. From there, you can make any changes that you want to the toolbar.



9. To change the setting where SPSS begins to look for files: Go to Edit

Options. Under the “File Folders” tab, search for the area from which you want SPSS to start looking for data files and other varieties of files.



10. To show colour coding with SPSS syntax: Go to Edit Options. Under the “Syntax” toolbar, at the top left corner, tap on the box that has “Display syntax colour coding.”

11. To turn on/off auto-complete for SPSS syntax: Go to Edit Options. Under the “Syntax " toolbar, in the top right corner, tap on the box that has “Automatically display the auto-complete control”. To turn off auto-complete, uncheck the box (see screenshot above).

12. To show the line numbers in the SPSS syntax window: Go to Edit Options. Under the “Syntax" toolbar, in the inner right box, tap on the box that has “Display line numbering” (see the above screenshot).

Properties of SPSS Variable View

In the Variable View, utilize the Role cell to determine the degree of measurement for the variable. Tap in the Role cell to initiate a drop-down bar list with six items and choose one:

It has:

Input: The variable is used as an input.

Target: The variable is used as a target variable.

Both: The variable will is used as both input and target.

Note: The variable has no function to perform

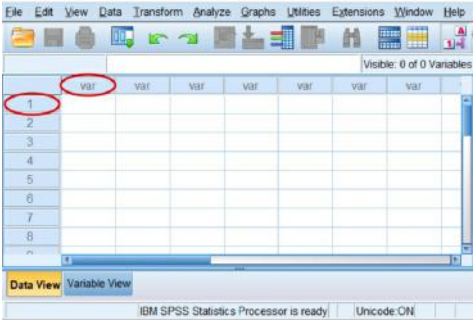
Partition: The variable separates the data into different samples for training, testing, and validation.

Split: The split function is Compatible with the IBM SPSS modeler. Variables with this function are not used as split file variables in IBM SPSS Statistics.

Chapter Two
Creating A New Data

When you launch the SPSS procedure, an empty spreadsheet in Data View will appear. To create a new one, tap on File > New > Data to open a blank spreadsheet.

You will see that each of the columns is named “var.” The column names will denote the variables that you type in your dataset. You will also discover that each row is called with a number (“1,” “2,” and so forth). The rows will portray cases that will be an aspect of your dataset. When you type in values for your data in the spreadsheet cells, each value will conform to a particular variable (column) and a particular case (row).

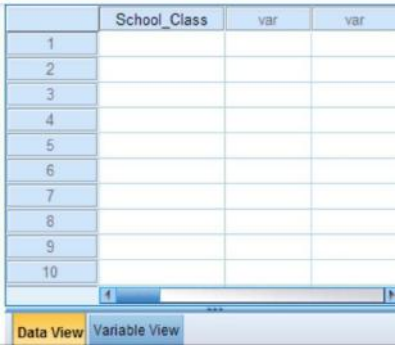


Below are steps to follow when entering data:

1. Tap on the Variable View tab. Insert the name for your first variable under the Name column. You can also insert other information about the variable, such as the type (the default is “numeric”), width, decimals, label, etc. Insert the name for each variable you intend to incorporate in your dataset. For instance, I will type “School_Class” as a variable for the class level of each learner (i.e., 1 = first year, 2 = second year, 3 = third year, and four = the fourth year). I will also indicate 0 decimals because my variable values will only comprise real

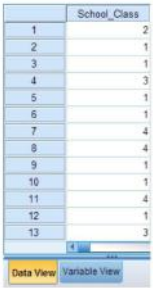
numbers. (The default is two decimals.)

2. Tap the Data View tab. Any variable names that you put in the Variable View will now show in the columns (one variable name per column). You will notice that School_Class emerges in the first column.



3. Type in numbers for every case. For instance, cases denote students. For every student, type a number for their class grade in the partition to the

suitable row and column. For example, the first student's data should appear in the first row, directly under the variable column School_Class. Here, in the example below, the first student's class grade is “2,” the second student’s is “1,” the third student’s is “1,” the fourth student's is “3,” and so forth.



	School_Class
1	2
2	1
3	1
4	3
5	1
6	1
7	4
8	4
9	1
10	1
11	4
12	1
13	3

Data View Variable View

4. Reiterate these steps for all variables included in your dataset. Click the save button to save your progress as you type in data systematically.

Importing Data

Before running any analysis in IBM SPSS Statistics software, you must first have data. You could type in data directly in the SPSS data view. Many times, we could have data from any other sources like data being stored in Microsoft Excel, or the information is in the form of text file such as CSV, txt,

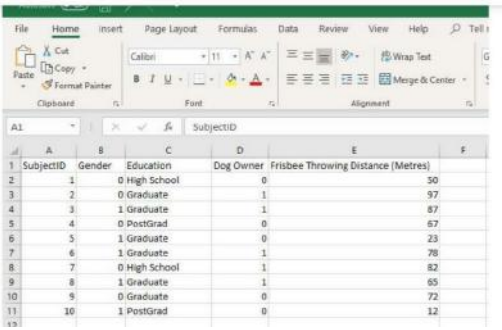
Importing Data From Excel

One of the most common ways to receive data is from Excel spreadsheets. Select demo. Excel. Double click. First, you will see that all the data are in sheet 1. Take note of the structure of the data. The numbers are organized in rows and columns. This is called Rectangular Data. You will also notice that the variable names are in the first columns as column headings. SPSS has strict rules on how you handle variable names. If the invariable names in your excel spreadsheet do not conform to SPSS rules on naming, they will be converted to valid SPSS variable names. The original variable names will be

saved as variable labels.

Getting started with Excel data

Click on the Excel spreadsheet you want to import your data from.

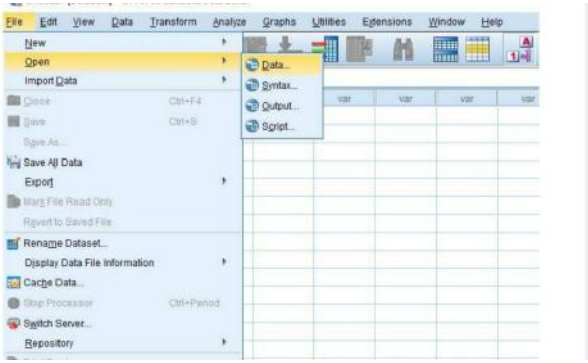


	A	B	C	D	E
1	SubjectID	Gender	Education	Dog Owner	Frisbee Throwing Distance (Metres)
2	1	0	High School	0	50
3	2	0	Graduate	1	97
4	3	1	Graduate	1	87
5	4	0	PostGrad	0	67
6	5	1	Graduate	0	23
7	6	1	Graduate	1	78
8	7	0	High School	1	82
9	8	1	Graduate	1	65
10	9	0	Graduate	0	72
11	10	1	PostGrad	0	12

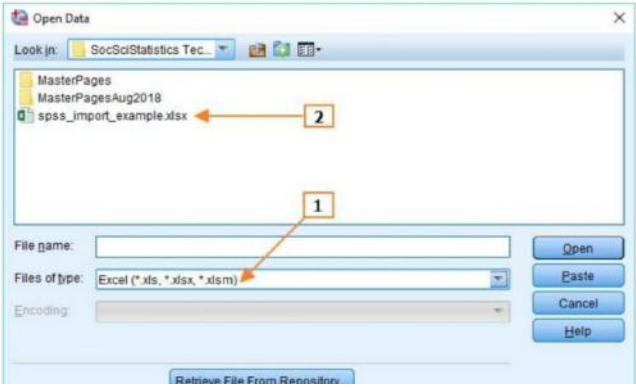
In this example, there are five columns of data.

Ensure your variable names are in the first row of the spreadsheet with no gap between the first and second rows. Also, please clean up the data, checking it cell by cell. Then you can start the procedure of importing your data into SPSS.

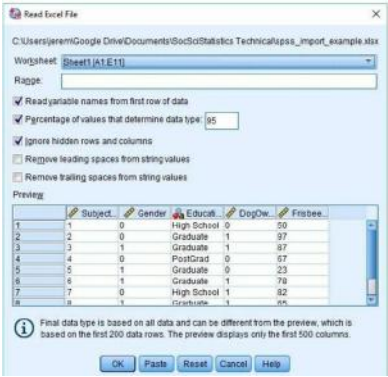
To begin, Tap File -> Open -> Data (See screenshot below).



Immediately you tap Data; the Open Databox will come out. (See screenshot below)

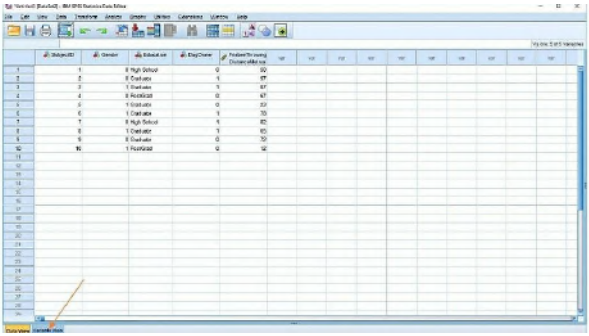


In the box with Files of type, choose Excel (see screenshot one above); then move to the folder that has your Excel file you want to import; your Excel file will pop up (see screenshot two above). Click on the file to open it; a Read Excel File dialog box will be displayed.



This dialog box gives numerous alternatives. This is how the new edition of SPSS selects a data variety to your variables. The percentage is the number of numbers in a specific column that must fit a particular data type for SPSS to allocate that data type to a variable. In the example, a variable will be assigned to a specific data type if 95% of values fit that type (for instance, if 95% is a number). The number will be above 50. . If the software fails to identify it, SPSS will convert it to a string variable. If your data looks good, click Ok to import the information,

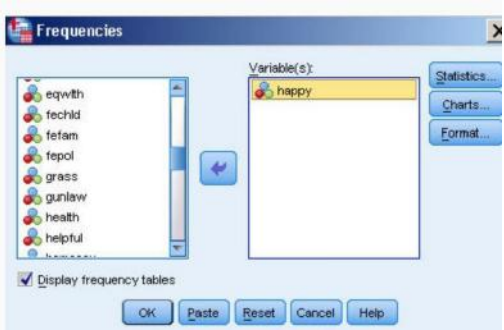
SPSS will bring your data into a new Data View with variable names at the first rows.



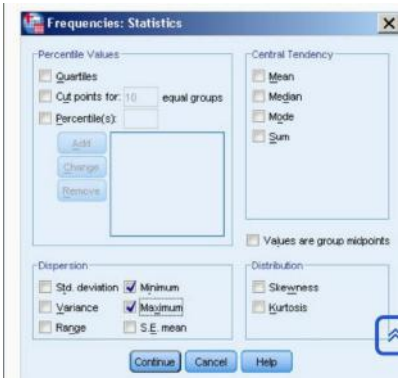
Ensure that the data transferred is successful. Then check that data parameter are correct for numerical data. Tap on the Variable View tab (see screenshot below). This will cause the variable view to appear within SPSS.

Cleaning Data For Import

When you are done typing in your data, you have to review them for mistakes. Then tap on the variable name you want to check and shift it to the Variable box. In the example below, I examine the variable “Happy” from the General Social Survey.



Tap on Statistics, and then Minimum and Maximum. Tap Continue and OK. See screen below.



This variable inquires the participant's general level of happiness:

The information entered should include only 0, 1, 2, 3, 8, and 9. I have changed the database so that there are numerous mistakes. There are precisely two errors. See the screenshot below.

Statistics

GENERAL HAPPINESS

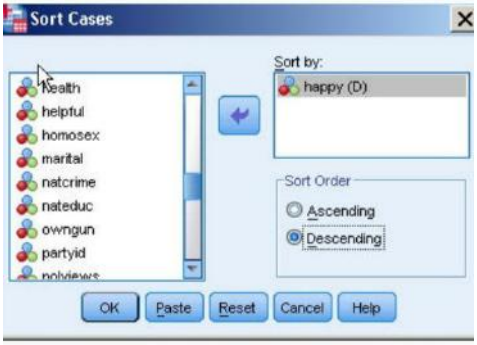
N	Valid	1495
	Missing	5
Minimum		1
Maximum		10

GENERAL HAPPINESS

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	VERY HAPPY	434	28.9	29.0	29.0
	PRETTY HAPPY	822	54.8	55.0	84.0
	NOT TOO HAPPY	237	15.8	15.9	99.9
	4	1	.1	.1	99.9
	10	1	.1	.1	100.0
	Total	1495	99.7	100.0	
Missing	DK	1	.1		
	NA	4	.3		
	Total	5	.3		
Total		1500	100.0		

To find the errors, sort the cases by either

arranging them in ascending or descending order. Tap on Data, Sort Cases. Tap on the name of the variable that has an error. (“happy”) and set it in the Sort By box. See the screenshot below.



Tap OK. Ensure you are in the Data View tab and not the output view tab. The cases with mistakes are close to the top of the lists (the ’10’ and the ‘4’). See screenshot below

If you do not have the data essential to recognize the case with the mistake, erase the value, and SPSS will attend to it as a missing

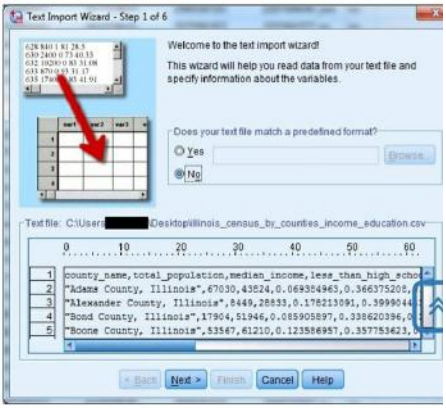
value.

Importing CSV File Using The Text import Icon

The dataset to be used will be in CSV file format. It means "comma-separated values." Files of this nature come with a ".csv" extension and are text files. They are supposed to be read as tables where every line stands for a row and columns, distinctly segregated by commas. SPSS can receive data from a CSV or any other text format using the command "File"->"Read Text Data..." selection. It will unlock a pop-up dialogue that enables the user to assign the CSV file from the computer. Immediately the file is set, the "Text Import Icon" will launch. The stages are as follows:

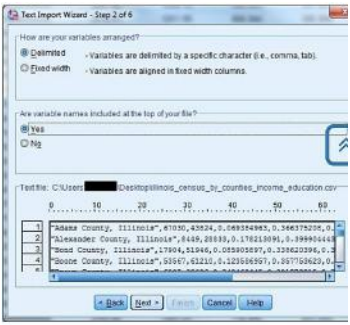
1. SPSS enables the user to save the settings utilized in importing the

CSV files. Stage1 wants to verify if you have saved the settings. For this example, tap "No." See screenshot below.

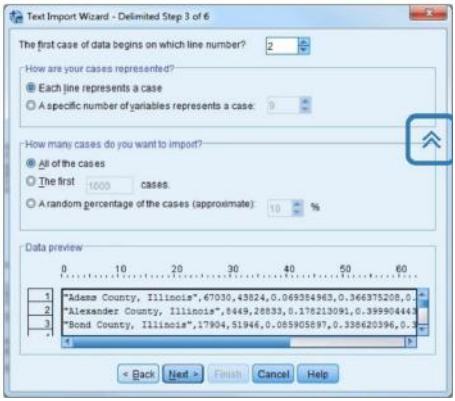


2. For stage 2, tap "Delimited" under "How are your variables arranged?" This notifies SPSS that the data has been structured in units allotted evenly by distinct symbols. Then click "Yes" in the box with "Are variable names included at the top of your file?" this stipulates that the first line of the file includes the column titles for the table that carries the

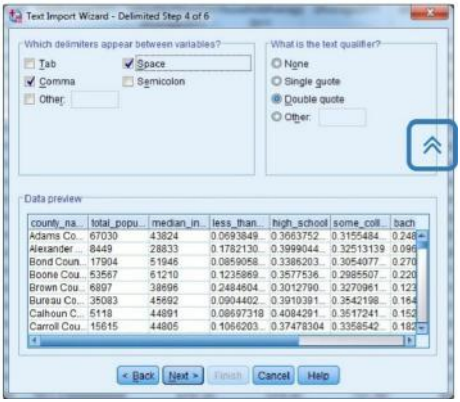
data. See the screenshot below.



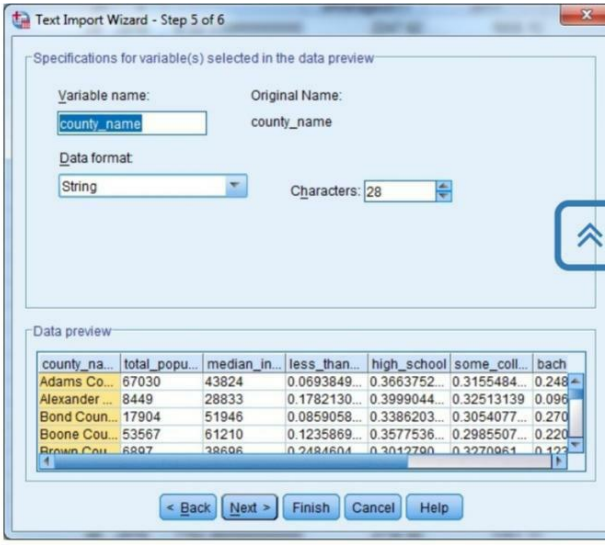
3. Stage 3 regulates which portions of the file will be dragged into SPSS. Since the column titles are on the first line, the first case will be set to "2." Each line denotes a case so that you will import all lines. See screenshot below



4. The box that says, "Which delimiters appear between variables?" Click "Comma" and "Space." "Delimiters" are the symbols that set apart each data value. Click on "Double Quote." See the screenshot below.



5. This stage enables you to make alterations to any of the variable data. The SPSS Data format of inconsistent restraints any value it is composed of: numbers are numeric, symbols are strings, and so forth. See the screenshot below.



6. Stage 6 enables the user to save the import format by clicking "Yes" in the box that says, "Would you like to save this file format for future use?"

The box that says"Would you like to paste the Syntax?" will publish the command syntax to the output viewer. The box that says "Cache Data Locally" will copy the information in memory to enhance performance. Unclick this for files more significant than one or two gigabytes according to your computer's memory. See screenshot.

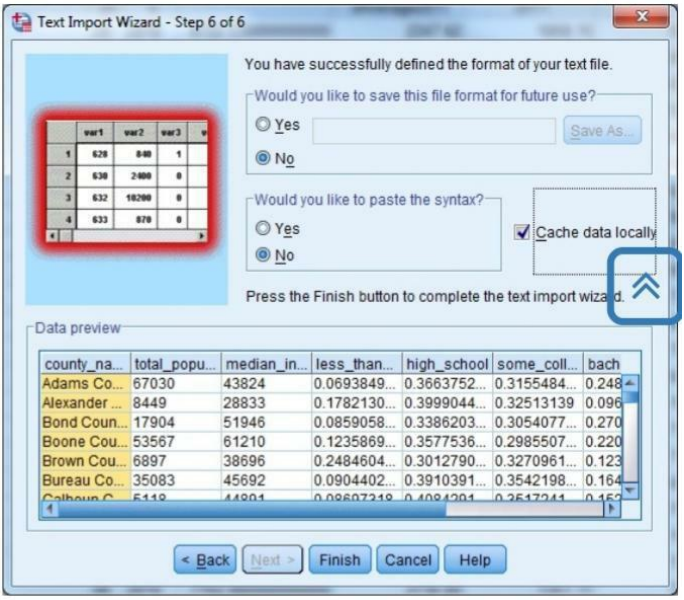


Chart Builder

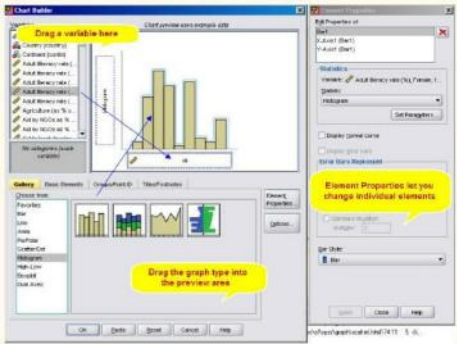
The SPSS Chart Builder is, as illustrated by its name, an interactive chart-building capacity conjured by Graphs > Chart builder. It allows you to formulate graphs by pulling and dropping several components into the chart preview region. See the screenshot below.

When you start the Chart Builder, an alarm dialog window opens on the screen, telling you everything about measurement statuses. If measurement statuses are announced as they are meant to be, you can overlook it; if they are not, carefully obey the alarm.

For traditional charts, all you have to do is:

1. Please choose one of the graph varieties shown in the Gallery (double-tap or pull it into the preview region).
2. Pull the variable(s) you intend to chart from the Variable sheet into distinct areas ("drop zones") that will show in the Preview region ("Canvas") after selecting a specific grape variety.

You can choose to amend several components of the graph utilizing one of the tabs or the properties window that pops up or leave it as it were. See the screenshot below.



Remark: The type of chart category chosen may cause some options not to be selectable. Several selections you can fix in the properties here can also be selected and altered in the Chart Editor after generating the chart.

Chapter Three
Analyzing Data

There are several types of data analyses accessible in SPSS, and they can be seen under the Analyze selection on the menu.

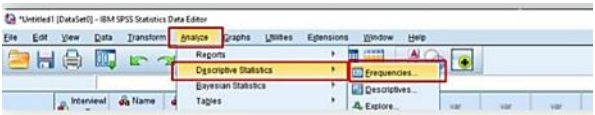
Descriptive Statistics For Categorical

Data

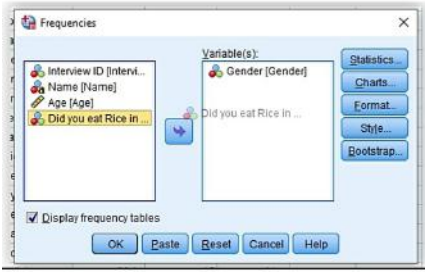
Descriptive statistics define a variable's central tendency (and dispersion (the distribution of the variable's responses).

Variables that were assigned as ordinal or nominal are analyzed accurately utilizing frequencies. For instance, if we want to know the number of males and females in a specific population.

1. Tap on the Analyze element on the menu, go to Descriptive statistics, and tap Frequencies. See screenshot below

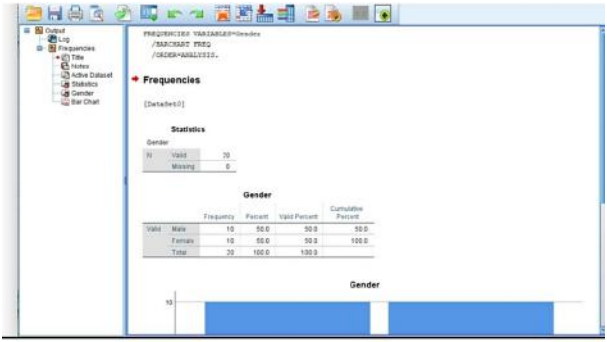


2. Go to the dialog box, tap, and pull the Gender variable into the box on your right. See the screenshot below.



3. Repeat the procedure for the “Did you eat Rice in the past seven days?”
4. Tap the Charts lever on the right of the dialog box.
5. Select bar charts and tap Continue

6. Tap OK.
7. An entirely different window known as the Viewer will emerge with the outcomes of the calculations.



The first table in the output is the Statistics. Since no statistics were assigned, there are only two valid and missing values. The Valid digit denotes the number of cases that have proper reactions on the variable.

The Missing digit denotes the number of cases that did not give a good reaction to the

variable view.

The second table symbolizes the Table of Frequencies.

Frequency depicts the number of times the value occurs in the data set.

Percent is the frequency implied as a percent out of the total number of cases in the data set

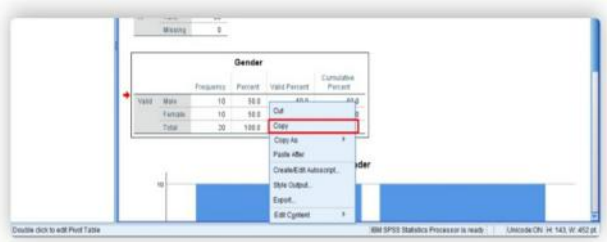
Good percent is the frequency conveyed as a percentage out of the Valid cases – the number of people who indeed reacted on the variable. This is the value that gets documented.

The cumulative percent is the total percentage of the species that has been calculated in that row; it can be evaluated by putting together all of the numbers in the Valid Percent column above the existing row.

To document the outcomes from the table, say:

There were 20 participants in the poll.10 were Male, and 10 were Female – each depicting 50% of the species, respectively.

To obtain the output to Microsoft Word for your documentation, right-tap on the production, for instance, a chart or a table, and assign a copy. In Microsoft Word, right-tap where you want the graph or table to appear and choose paste.



Frequency Command Menu

The frequency command is used In creating fundamental charts like histograms and bar graphs. The Frequency command menu gives the percentiles and other statistics. Below are the controls found in the menu:

1. **The sort order command:** SPSS default setting for sort order can be altered by adding a subcommand.

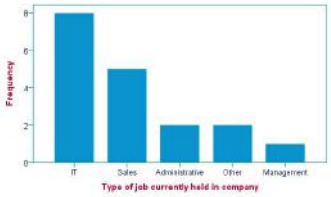
Apparent VALUE and VALUE (ascending and descending values) or AFREQ and DFREQ (ascending and descending frequencies).

2. **Bar Chart Command Menu:**

This frequency command is the simplest way to generate numerous bar charts for categorical variables. You can put it in the bar chart subcommand.

Remark:

You can incorporate this command with sort order, making the bar chart bars ordered in ascending order. See the screenshot below.



3. The Percentiles gives a pleasant manner of achieving percentiles. Upload a percentile subcommand accompanied with the preferred

percentiles in parentheses. Note that percentiles are not significant for nominal variables.

4. **The Ntile:** Niles is effortlessly received with the SPSS Frequency command and uploaded to the NTILES subcommand with various tiles in the back of it in parentheses. If you need to assign cases to tile groups, use RANK; it generates a new variable retaining the tile for every point on a given variable. Both alternatives are proven in the syntax. Rank monthly_income/ tile.

See the screenshot below.

Statistics

Gross monthly income in Euros (excluding bonuses)

N	Valid	19
	Missing	1
Percentiles	20	2116.32
	40	2409.22
	60	2854.31
	80	3228.90

5. **Statistics**

SPSS Frequency command: This can evaluate all calculations acquired from descriptive, including the median and mode. Note that the statistics from frequencies have a specific format with variables in columns and statistics in rows. To compile them, upload a statistics subcommand. The same as with descriptive stipulating the ALL key-word returns all accessible statistics.

Acquire all accessible statistics.

Frequencies monthly_income/ format notable/ all statistics.

See the screenshot below.

Statistics

Gross monthly income in Euros (excluding bonuses)

N	Valid	19
	Missing	1
Mean		2660.31
Std. Error of Mean		128.311
Median		2546.47
Mode		1821[a]
Std. Deviation		559.293
Variance		312808.402
Skewness		.432
Std. Error of Skewness		.524
Kurtosis		-.648
Std. Error of Kurtosis		1.014
Range		2011
Minimum		1821

6. SPSS frequency Multiple Variables command d: Obviously, Frequencies

may be run for a couple of variables, probably, the usage of TO or ALL. If a couple of output styles (frequency table, chart, and so forth) are created, you may have them taken care of through variable or type of output through specifying VARIABLE or ANALYSIS on an ORDER subcommand. Format free bar chart/order analysis.

Descriptive Statistics For Two categorical Variables (Chi-Square)

The Chi-Square Test of Independence is utilized to examine if two categorical variables are correlated.

The Chi-Square Test of Independence determines whether or not there may be an

affiliation among categorical variables (i.e., whether or not the variables are independent or related). It is a multivariate check.

This check is called the Chi-Square Test of Relation.

This check makes use of a contingency table to evaluate the information. A contingency table (also called a cross-tabulation or cross tab) is an array wherein information is classified according to specific variables. The classes for one variable are seen in the rows, and the types for the alternative variable appear in columns. Each variable ought to have two or more categories. Each cell expresses the entire count of cases for a particular type of pair.

Data Requirements

Your information needs to fulfill the following requirements:

1. Two specific variables.
2. Two or more categories (groups) for every variable.

3. Independence of compliances.
4. There is no association among the topics in every group.
5. The distinct variables are not "paired" in any way (e.g., pre-check/post-check observations.
6. Significantly large structure size.
7. Predicted frequencies for every cell are at least 1.
8. Predicted frequencies must be at least five for the prevalence of the cells.

The Theorem

The void theorem (H0) and alternative theorem (H1) of the Chi-Square Test of Independence may be expressed in two distinct but equal ways:

H0: "[Variable 1] is impartial to [Variable 2]."

H1: "[Variable 1] isn't impartial to [Variable 2]."

OR

H0: "[Variable 1] isn't related to [Variable 2]."

H1: "[Variable 1] is related to [Variable 2]."

The Statistics Check

The statistic check for the Chi-Square Test of impartial is indicated as Y^2 and is evaluated as:

Y2=∑i=1R∑j=1C(or−eij)^2eij

where

oil is the observed cell count in the ith row and jth column of the table

it is the predicted cell count in the ith row and jth column of the table, calculated as

eij=(row I total∗col j total)/(total)

The amount of (or - eij) is occasionally known as the residual of the cell (i, j), indicated by rij.

The calculated Y^2 rate is then described as

the actual speed from the Y^2 distribution table with ranges of freedom df = (R - 1)(C - 1) and selected reliance level. If the calculated Y^2 rate > actual Y^2 rate, then we dismiss the void theorem.

Setting Up the Data For Analysis

There are two distinctive methods wherein your statistics can be installed originally. The layout of the statistics will specify a way to continue with running the Chi-Square Test of impartial. At a minimum, your statistics must consist of specific variables (denoted in columns) to be used in the analysis. The particular variables need to consist of at least groups. Your statistics can be formatted in either of the following methods:

Raw data like this shows that every row is a topic. See the screenshot below.

	ids	Smoking	Gender
1	20183	Nonsmoker	Male
2	20230	Nonsmoker	Male
3	20243	Past smoker	Female
4	20248	Current sm...	.
5	20255	Nonsmoker	Female
⋮			
430	49821	Past smoker	Female
431	49838	Nonsmoker	Male
432	49854		Male
433	49879	Nonsmoker	Male
434	49931	Nonsmoker	Male
435	49947	Nonsmoker	Female

Cases stand for topics, and every once in the dataset. That is, each row stands for a remark from a distinct subject.

The dataset includes two nominal categorical variables (string or numeric). The categorical variables employed in the investigation must have two or more categories.

If there are frequencies (every row is a mixture of elements)

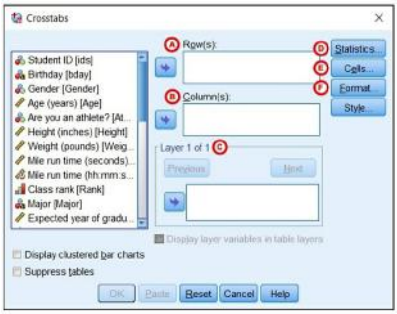
	ClassRank	PickedAMajor	Freq
1	Freshman	No	212
2	Freshman	Yes	114
3	Sophomore	No	171
4	Sophomore	Yes	168
5	Junior	No	92
6	Junior	Yes	198

Before running the investigation, you must initiate Weight Cases and put the frequency variable as the weight.

Perform a Chi-Square Check of Impartial

In SPSS, the Chi-Square Check of Impartial is selected within the Crosstabs technique.

To generate a crosstab and execute a chi-square check of impartial, tap Analyze > Descriptive Statistics > Crosstabs.



A Row(s): One or more variables to apply in the rows of the crosstab(s). You need to input as a minimum one Row variable.

B Column(s): One or more variables to apply in the crosstab columns (s). You ought to input as a minimum one Column variable.

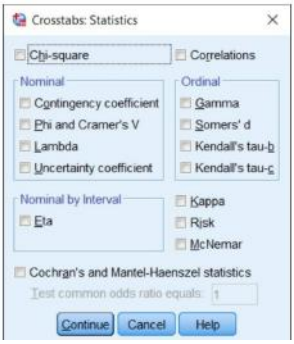
Also, notice that if you define a one-row

variable and or more column variables, SPSS will publish crosstabs for every pairing of the row variable with the column variables. The same is confirmed when you have one column variable and or extra row variables or when you have more than one row and column variables. A chi-rectangular check could be produced for every table. Moreover, if it consists of a layer variable, chi-rectangular checks could be run for every pair of row and column variables within every stage of the layer variable.

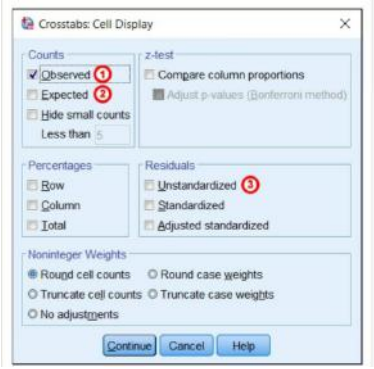
C Layer: A discretionary "stratification" variable. The chi-rectangular check outcomes feature a distinct layer variable, SPSS will subset the facts with regards to the type of the layer variable, then run chi-rectangular checks among the row and column variables. (This isn't equal to checking out a three-way correlation or checking out for a correlation among the row and column variables after regulating the layer variable.)

D Statistics: Opens the Crosstabs: Statistics window, which incorporates fifteen particular

inferential records for evaluating specific variables.



Ensure that the Chi-Square box is ticked before running the procedure.



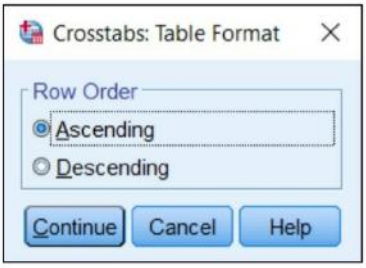
1. Observed: The entire range of remarks for an allotted cell. This alternative is allowed through default.

2. Expected: The predicted range of remarks

for that cell

3. Unstandardized Residuals: The "residual" value, evaluated as discovered minus predicted.

F Format: Opens the Crosstabs: Table Format window, which indicates the way the rows of the table are ordered.



Recoding Variables

Recoding can be used to generate various values or codes for a Variable.

Recoding into various variables is achieved using three kinds of recording, namely:

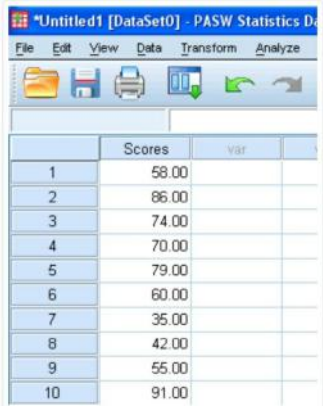
1. Recoding an assigned set of values.
2. Recoding a unit variable.
3. Recoding data into two groups

1. Recoding Assigned Set Of Values

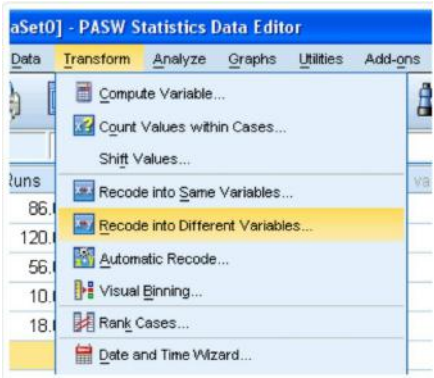
Example: The information provided below illustrates the ratings of 10 college students in the very last examination. Recode the facts bestowing code "1" to ratings among 75 - 100, code 2 to ratings between 61 - 74, code three to ratings among 41 - 60, and code four to ratings among 0 - 40.

Scores	58	86	74	70	79	60	35	42	55	9

A. Type in the information in the SPSS Statistics Data Rewriter and call the variable "Scores."Remember that every person's outcomes go on a distinct line (row) in SPSS Statistics.

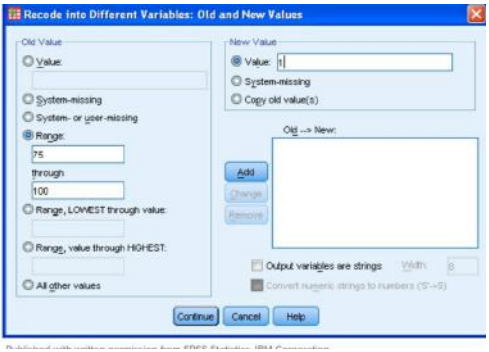


B. Tap on Transform > Recodes Into Different Variable at the toolbar.

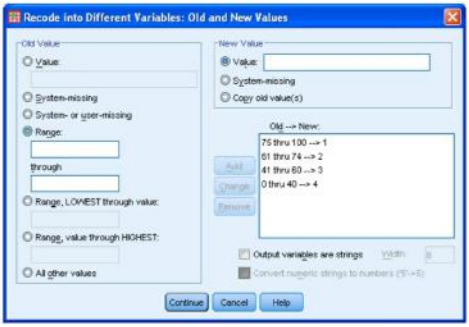


C. Substitute the variable you need to recode by assigning it and tapping the SPSS Arrow lever on the right, and bestow the unique variable a title

and tag. In the example, we have given the different variable a title of "scores." See the screenshot below.

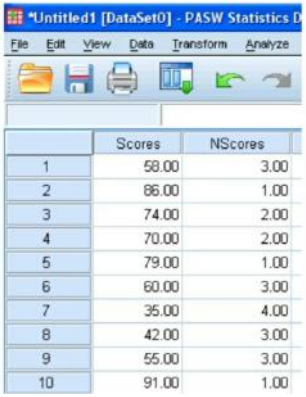


D. Tap on the SPSS Change lever.
E. Tap on the SPSS box that says Old and New Values.
F. Type the first set of "75 - 100" into the set, but the new code to "1" into the Value.



G. Tap on the SPSS Adds up lever.

H. Redo the process for remaining values, and your screen will look like the screenshot below.



	Scores	NScores
1	58.00	3.00
2	86.00	1.00
3	74.00	2.00
4	70.00	2.00
5	79.00	1.00
6	60.00	3.00
7	35.00	4.00
8	42.00	3.00
9	55.00	3.00
10	91.00	1.00

2. Recoding A Unit Variable

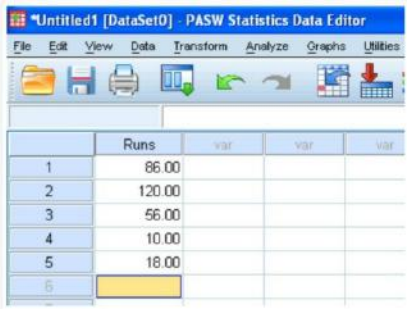
Example: The records below denote races achieved by five batsmen in a national-

competition match. Recode the documents in such a way that the batsmen are ranked by their quantity of runs, with the batsman with the best runs bestowed a code of "1" and the batsman with the least runs bestowed a "five."

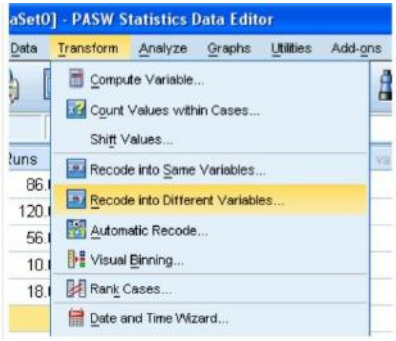
	Number of runs by batsmen				
Batsmen	1	2	3	4	5
Runs	86	120	56	10	18

A. Type in the records in the SPSS Statistics Data Rewriter and call the variable "Runs."

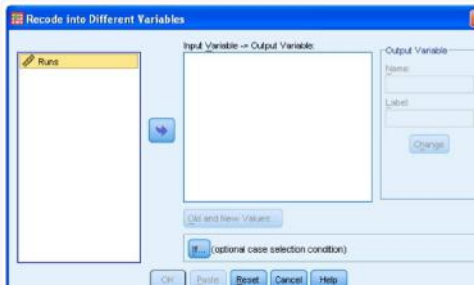
Note that every individual's outcome is entered on a different line (row) in SPSS Statistics.



B. Tap on Transform > Recodes Into Different Variable at the toolbar.

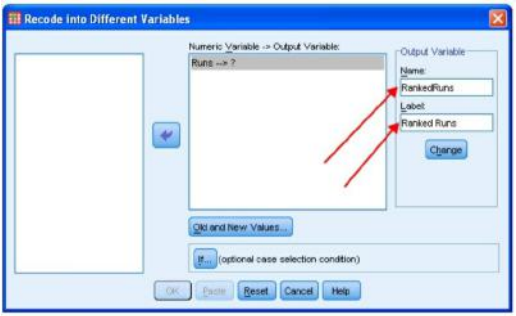


C. Intensify the Runs variable in the left-hand screen by tapping on it and then tap the SPSS Right Arrow lever to push it across to the Numeric Variable -> Output Variable: box

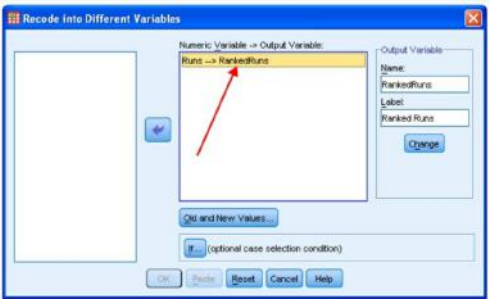


D. In the –Output Variable– region

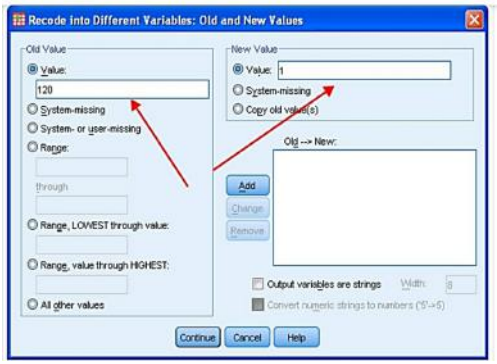
indicated below, provide the unique variable you generate a title and tag. In this example, we have called the new variable "Ranked-Runs" and the label "Ranked Runs."



E. Tap on the box that says 'change.'
F. You will notice that the Number Variable -> Output Variable: box has been altered to reflect these changes.

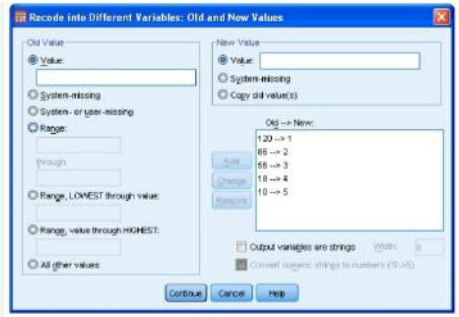


G. Tap on the box that says old and new values.
H. Type the first score into the Value and put the new code of "1" into Value.
I. Tap on the box that says add.
J. Redo the process for the remaining values. See the screenshot below.



K. Tap on the box that says Continue.
L. This will take you back to the former screen and tap on the SPSS OK. This will bring you to the Data View Rewriter, which will now indicate that you have generated a unique variable that is based on recoding

your existing occurring.



3. Recoding Data Into two Groups.

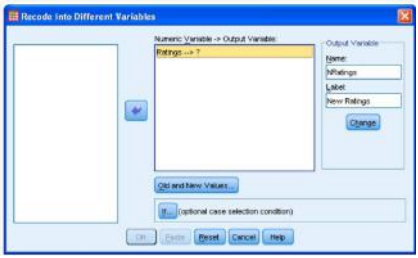
Example: The records provided below denote an achievement grade out of 10 for an exotic service offered by a firm. The firm would like to code everyone who replied, giving ratings above five a "Satisfactory" code and those below five a "Dissatisfactory" code.

Scores	3	6	8	9	7

A. Type in the information in the SPSS Statistics Data Rewriter and call the variable "Ratings."

B. Tap on Transform > Recodes Into Different Variable at the top of the toolbar.

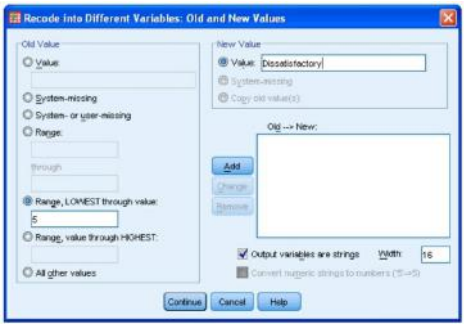
C. Move the variable you need to recode by choosing it and clicking the SPSS Arrow lever on the right, and name and tag the new variable. See the screenshot below.



D. Tap on the SPSS Change lever.

E. Tap on the SPSS box that says Old and New Values.

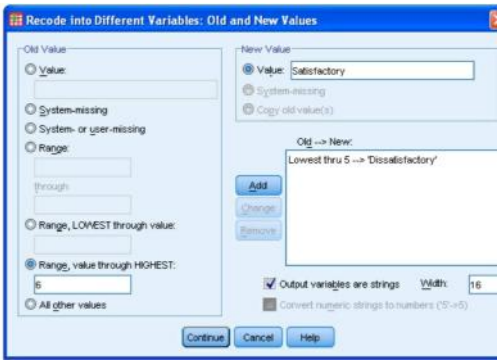
F. Type the value of "5" into the scale, most minor through value, and put the new code to "1" into the Value. Tap the box that says ' Output variables are strings' to receive values that are not numeric. You need to expand the Width: from 8 to 16 so that the string tags are not shortened.



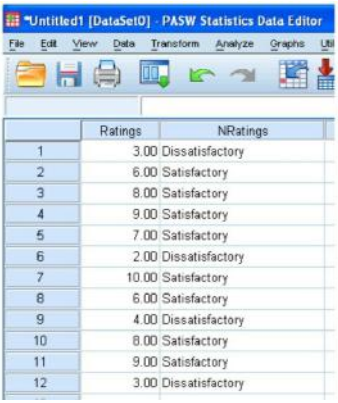
G. Tap on the SPSS Arrow lever on the right.

H. Type the value of "6" into the scale value through the enormous box and

put the new code to "1" into the Value-region.



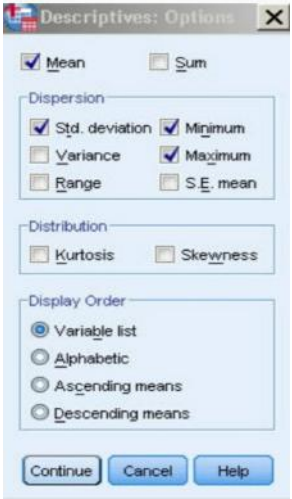
I. Tap on the SPSS Add lever.
J. Tap on the Continue lever.
K. This will take you back to the former screen. Tap on the SPSS OK, and it will bring you to the Data View Rewriter. This will indicate that you have generated a new variable based on recoding your occurring variable.



	Ratings	NRatings
1	3.00	Dissatisfactory
2	6.00	Satisfactory
3	8.00	Satisfactory
4	9.00	Satisfactory
5	7.00	Satisfactory
6	2.00	Dissatisfactory
7	10.00	Satisfactory
8	6.00	Satisfactory
9	4.00	Dissatisfactory
10	8.00	Satisfactory
11	9.00	Satisfactory
12	3.00	Dissatisfactory

Descriptive Statistics For Scale Data

To get descriptive statistics from continuous variables, tap Analyze, Descriptive Statistics, Descriptives. Navigate your variables into the Variable box. Tap on Options and make the following choices:



Tap Continue, OK.

Visual Binning

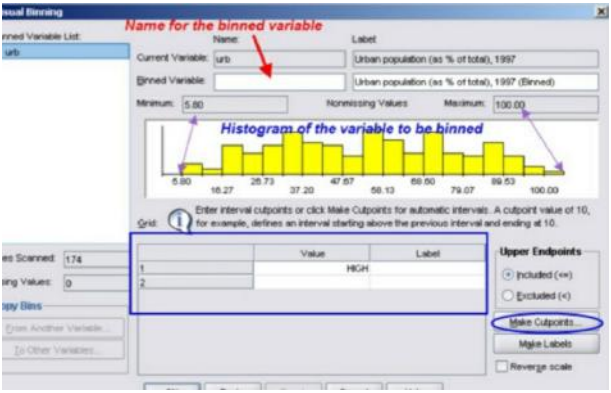
Visual binning allows you interactively build groups (bins, groups) from a fixed variable and visually regulate the procedure.

The example below "bins" urb (% of persons residing in cities in several countries, i.e., builds a categorical variable from the data.

Transform > Visual Binning.

On the first dialog box, define the variable(s)

you need to recode ("bin")



Select a title for the Binned variable; SPSS already provides a default Tag, but you can change it if you wish.

Type in the (Grid) cutpoints, i.e., the (upper) thresholds that separate two periods (bins) and the interrelated tag. (HIGH means the highest value, in our example, a matter of 100.0).

If you want to divide the variable regularly, utilize Make Cutpoints.

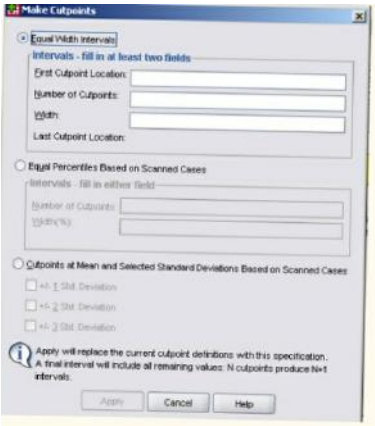
Proportional Width periods define evenly spaced periods. Type in either the width or the number of periods.

Proportional percentiles on examined cases define groups/classes with approximately the same observations. Put in the number of cut points or the %width by typing the percentage of patients in each group. For instance, if you indicate three cut points or 25% of the cases, you will build four categories comprising one-fourth of the observations.

Cutpoints at mean/standard deviations:

Stipulate a cut point at the mean and ± 1,2 or 3 standard deviations.

Then click Apply



Select a name for the Binned variable; SPSS already provides a default Label, but you can

change it if you wish.

Type in the (Grid) cutpoints, i.e., the (upper) limits that divide two periods (bins) and the matching label. (HIGH implies the highest value, in the above example, the highest value is 100.0).

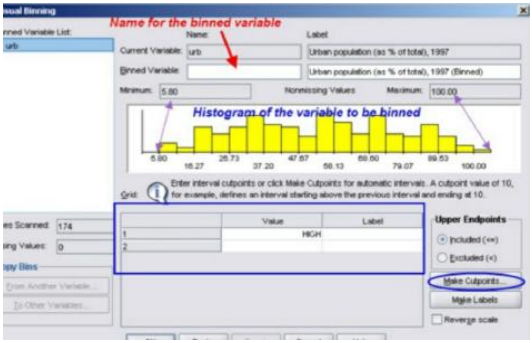
If you need to divide the variable, utilize the option box that says Make Cutpoints:

Proportional Width periods interpret uniform spaced periods.

Proportional percentiles on examined cases define classes with approximately equal observations. Put in either the number of cut points or the %width by typing the percentage of patients in each category. For instance, if you indicate three cut points or 25% of the cases, you will build four classes incorporating each one-fourth of the observations.

Cutpoints at median/standard deviations: Establish a cut point at the median and ± 1,2 or 3 standard deviations.

Then click Apply.

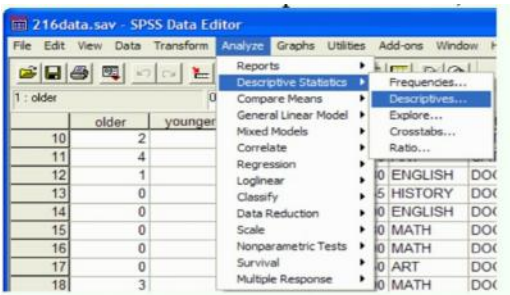


The Visual Binning dashboard will be displayed: the bins (periods borders) have been entered by the regulations you indicated. Remark: HIGH means up to the most significant value of the variable.

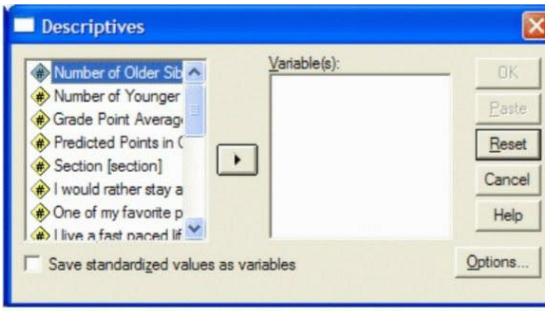
Descriptive Command Menu

The descriptive command can be employed to infer estimates of central tendency (average), estimations of dispersion (range, standard deviation, variance, minimum and maximum), and forecasts of kurtosis and skewness. The command is located at Analyze | Descriptive Statistics | Descriptives (shorthand for tapping on the Analyze menu

icons at the top of the window and then tapping on Descriptive Statistics from the pull-down menu. See the screenshot below of Descriptives from the pop-up menu.

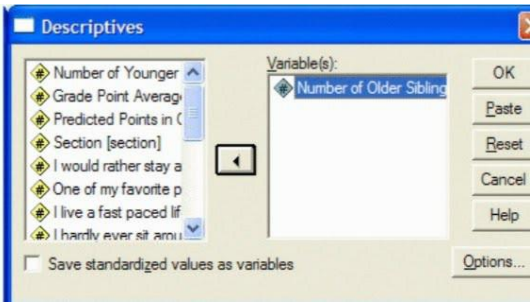


The descriptive dialog box will come up here.

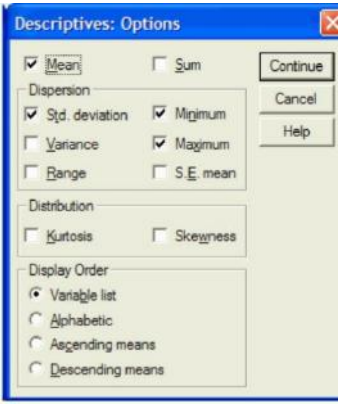


Choose the variable(s) that you need to evaluate by tapping on it in the left-hand sheet of the descriptives dialog box. Then tap on the arrow lever to shift the variable into the Variables sheet.

See the screenshot below.



Indicate which statistics you need to conduct by tapping on the Options lever. The Options dialog box will be displayed. See the screenshot below.



Choose the statistics you need by tapping on them (e.g., average, standard deviation, variance, range, minimum, etc.). Then tap on the Continue lever. Tap on the OK lever in the Descriptives dialog box. The SPSS Output Viewer will be displayed with the

outcomes in it.

Create Z scores

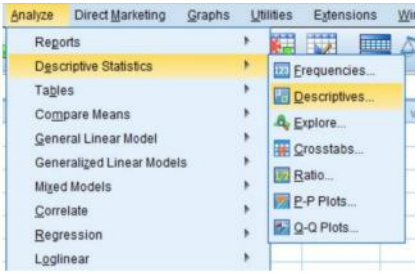
A Z-score denotes the number of standard deviations (SDs) a data point is away from the group's mean. It is popularly called a standard Z-scores. Hence, they are a valuable way of homogenizing values.

To create and evaluate Z- scores. I will use a random list of 10 scores in a test.

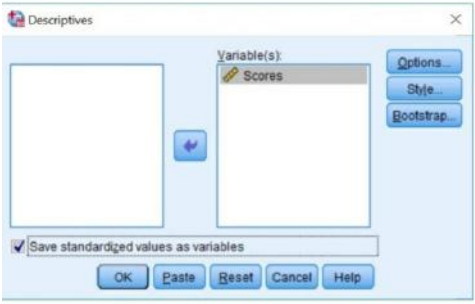
See screenshot below

	Scores	var
1	2	
2	5	
3	6	
4	5	
5	8	
6	3	
7	7	
8	5	
9	3	
10	1	

1. To evaluate the Z-scores, move to the Descriptives and tap Analyze > Descriptive Statistics > Descriptives.



2. Go to the scores required to be modified into the Variable(s) box to the right. Tap the selection that says Save the standardized values as variables. Mark the box which allows you to generate Z-scores.



3. Lastly, tap the OK lever.

When you move to the Data View in SPSS, you will notice a new variable has

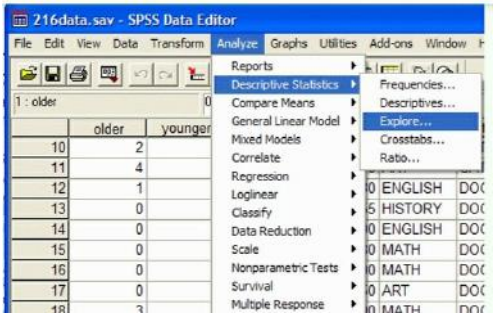
been formulated, which is labeled as the same variable as earlier with a Z prefix attached to it. See the screenshot below.

Scores	ZScores
2	-1.12
5	.22
6	.67
5	.22
8	1.57
3	-.67
7	1.12
5	.22
3	-.67
1	-1.57

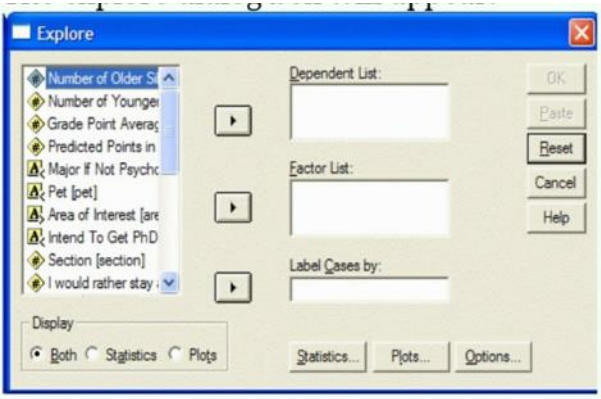
Explore command Menu

The explore command can be utilized to deduce estimates of central tendency (average and median), estimates of dispersion (range, interquartile range, standard deviation, variance, minimum and maximum), calculations of kurtosis and skewness, and formulate histograms, stem and leaf plots, and Tukey box plots. The command is

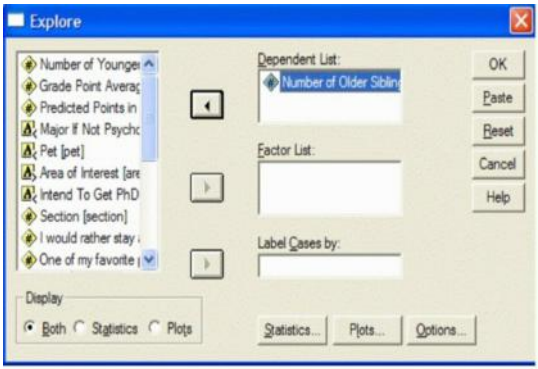
created by typing Analyze | Descriptive Statistics | Explore.



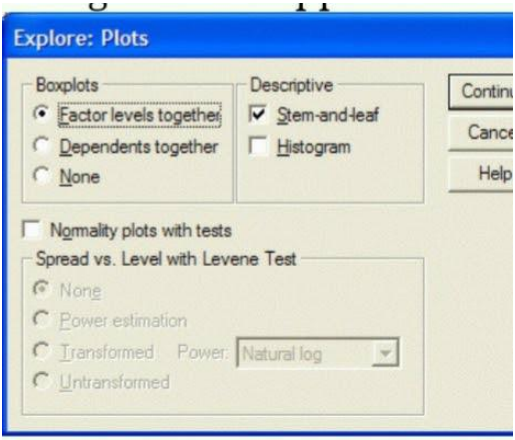
The explore dialog box will be displayed, as seen in the screenshot below.



Choose the variable(s) that you need to evaluate by tapping on it in the left-hand sheet of the explore dialog box. Then tap on the top arrow lever to shift the variable into the Dependent List.



Indicate which plots you need to assemble by tapping on the Plots lever. The Plots dialog box will be displayed, as seen in the screenshot below.



Determine the plots that you need by tapping on them. Then tap on the Continue lever. Tap on the OK lever in the Explore dialog box.

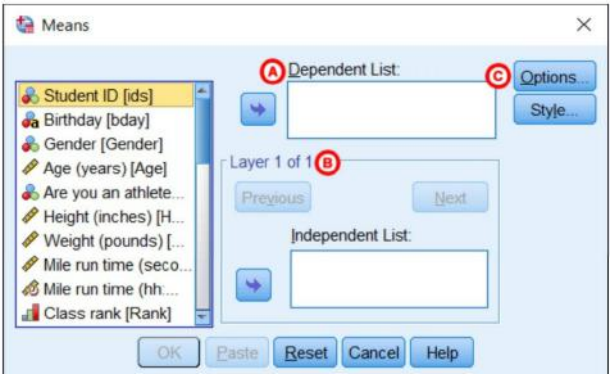
The SPSS Output Viewer will display your outcomes in it.

Comparison with Compare Mean Command

Compare Mean is used to differentiate numerous numeric variables concerning one or more specific variables. It is specifically valuable for summarizing numeric variables concurrently throughout the groups.

The Compare Means technique is beneficial when you summarize and evaluate variations in descriptive records throughout one or more components or specific variables.

To access the Compare Means technique, tap on Analyze > Compare Means > Means.

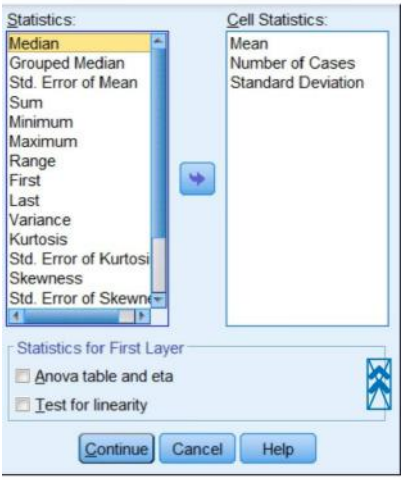


A Dependent List: This is the successive numeric variables to be analyzed. You must input at least one variable in this box before performing the Compare Means technique.

B Independent List: The specific variable(s) utilized to subset the based variables. Defining more than one value in the "Layer 1 of 1" box will generate numerous tables, each layer variable. You can define multiple layers for one table by tapping Next and then filling into different distinct variables; this can create a table that seems like a combination of a crosstab and the Descriptive technique.

C Options: This feature unlocks the Means selection window, where you could indicate the summary records to create and what

arrangement they must be indexed in.



The Statistics column at the left indicates what records are accessible. Overview records accessible include: average, the number of cases, preferred deviation, median, grouped median, liked mistakes of average, total, minimum, maximum, range, first, last, variance, kurtosis, chose mistakes of kurtosis, skewness, widespread errors of skewness, harmonic average, geometric average, percentage of the overall total, and percentage of available N. The Cell Statistics column on the right are the information that will be generated in the output. By default,

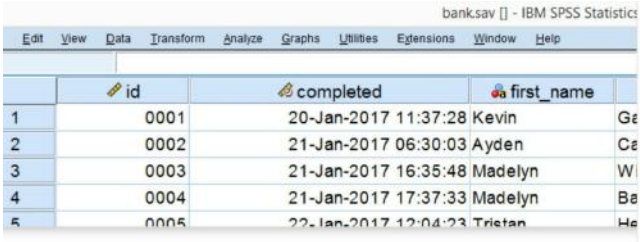
the average, quantity of cases, and preferred deviation could be evaluated. You can upload more information by tapping and pulling them from the Statistics column to the Cell Statistics column. You can also tap on and remove objects in the Cell Statistics column to alter the arrangement they occur in the output.

The Statistics for First Layer region consists of alternatives that will carry out one-way ANOVA and evaluate linear fit information (R, R2, Eta, and Eta Squared), respectively.

Chapter Four
THE OUTPUT

Basic Syntax

Consider the following table below as an example of data to be used.



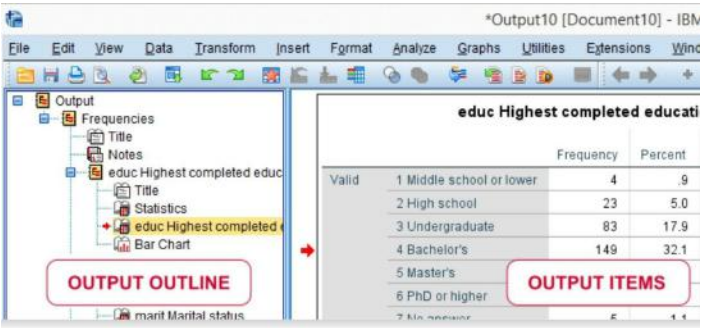
	id	completed	first_name	
1	0001	20-Jan-2017 11:37:28	Kevin	Ga
2	0002	21-Jan-2017 06:30:03	Ayden	Ca
3	0003	21-Jan-2017 16:35:48	Madelyn	W
4	0004	21-Jan-2017 17:37:33	Madelyn	Ba
5	0005	22-Jan-2017 12:04:23	Tristan	He

The basic syntax used in running this output in SPSS very below.

```
*Run frequency tables and bar charts on 3 variables.

frequencies educ marit jtype
/barchart
/order variable.
```

Performing this syntax opens an output viewer window, as seen in the screenshot below.



As demonstrated, the SPSS output viewer window usually has two primary panes:

the output outline is mainly used for moving through your output objects and

the real output objects -especially tables and charts- are always exported to WORD or Excel for documenting.

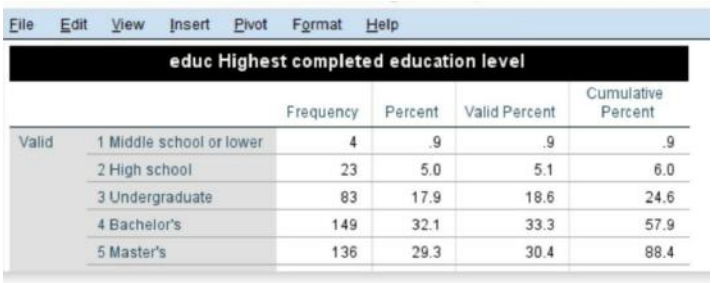
In the output outline, you could also erase output objects -SPSS always generates greater output than you expect. Use the ctrl button to choose numerous things. A faster way for erasing an option of output objects is OUTPUT MODIFY.

You can also decline and rearrange output objects in the outline. The most crucial ones are tables and charts.

SPSS Output - Tables:

If you need to make a few modifications to your output tables, one alternative is right-tapping the table and choosing Edit content SPSS Toolbar Arrow In a different Window as illustrated below.

The pivot table editor window (demonstrated below) enables you to modify anything about your table.



File Edit View Insert Pivot Format Help

educ Highest completed education level

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1 Middle school or lower	4	.9	.9	.9
	2 High school	23	5.0	5.1	6.0
	3 Undergraduate	83	17.9	18.6	24.6
	4 Bachelor's	149	32.1	33.3	57.9
	5 Master's	136	29.3	30.4	88.4

SPSS Pivot Table Editor Screenshot

I suggest you utilize the pivot table editor if everything else ceases to function.

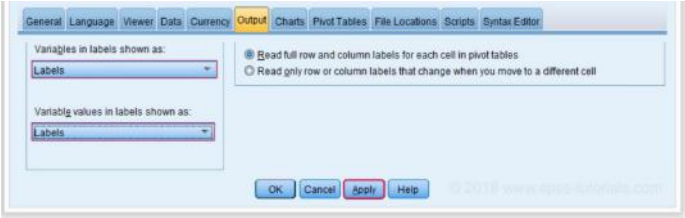
The reason is that you cannot duplicate and rerun anything you do in the pivot table editor. And more importantly, there are faster alternatives with which you may modify many tables in a single go.

Variable and Value Tags in Output Tables

When analyzing my data, I need to see variable titles and tags in my output. The same goes for values and value labels because I need to understand the number of my variables that have been coded.

Yet, I need to see only tags in the last tables that I'll document. One way to do this is to go to Edit SPSS toolbar Arrow Selections and choose the Output tab.

SPSS - Show Tags in Output from Options Toolbar



Strangely, the alternatives dialog has no Paste key. This is because it generates very sloppy syntax. A vastly higher alternative for this than the crappy toolbar is simply running the syntax below.

```
'Show only value labels and variable labels in output tables
set
numbers labels
vars labels.
```

TVARS is the short name for ”table variables'. ' It establishes how variables are exhibited in tables: titles, tags, or both.

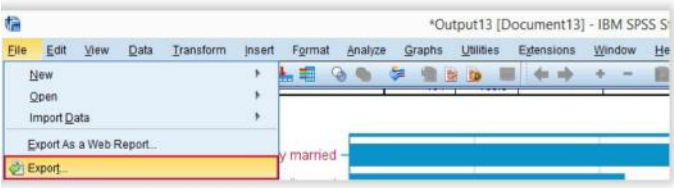
TNUMBERS is the short name for “table numbers.” It establishes how values are displayed in tables: values, tags, or both.

Performing this syntax is a much higher alternative than using the previous toolbar.

SPSS - All Output to MS WORD

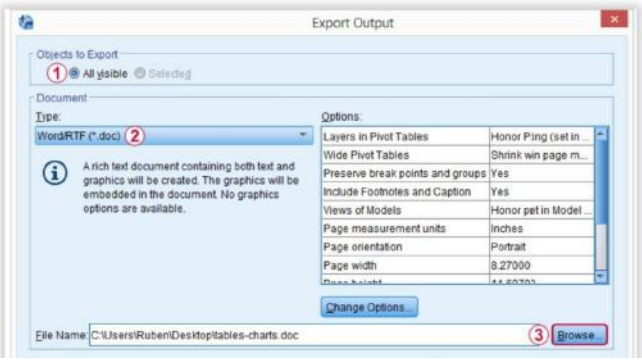
An outstanding manner to transform SPSS

output to WORD is exporting all contents of the output viewer in a single go. You can achieve this by going to File SPSS Toolbar Arrow Export as displayed underneath.



SPSS Output Export from Menu

You assign the alternatives below in a WORD file containing all tables and charts exhibited in your output viewer. This is the precise starting line for writing your statement.

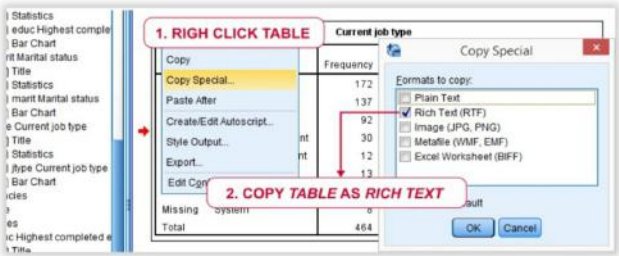


SPSS Output Export To MS WORD

Since you can edit tables and charts in SPSS - however, not in WORD- ensure your output is as suitable as it gives before changing it. An outstanding way to modify tables and erase unwanted output objects is OUTPUT MODIFY.

Copy-Pasting SPSS Output Tables to WORD

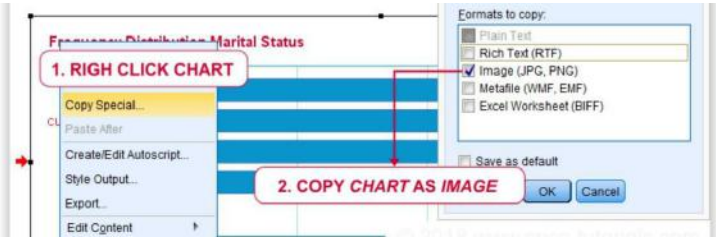
If you want only a few output objects in WORD, you may copy-paste them. The ctrl + c and ctrl + v shortcuts typically operate excellently for each table and chart. If it fails, however, use a copy unique for tables as indicated beneath.



SPSS Output Table - Copy Special To WORD

The same is going for copy-pasting charts to WORD: first attempt ctrl + c. If that doesn't

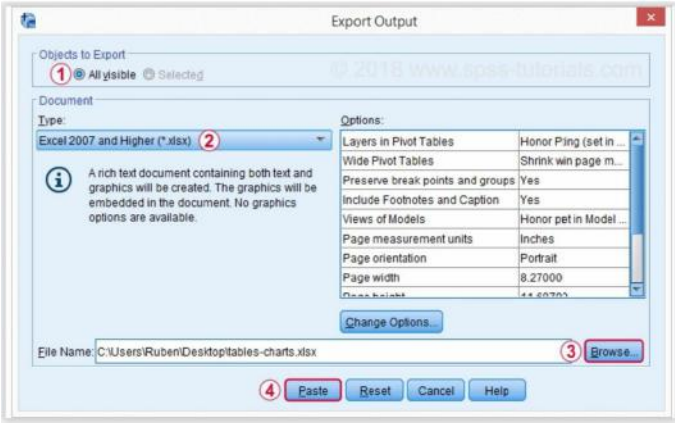
work, copy the chart as a picture as indicated below.



Output Chart Copy Unique To WORD

SPSS - All Output to Excel

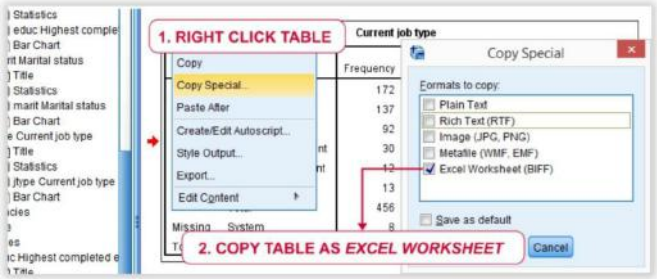
You can transform all contents of your output window -which includes all tables and charts- in one go to a single Excel sheet. To do this, go to File SPSS Toolbar Arrow Export and choose the alternatives illustrated below.



Output Export To Excel in SPSS

Copy-Pasting SPSS Output Tables to Excel

If you want numerous output tables in Excel, you may copy-paste them with the ctrl + c and ctrl + v shortcuts. If that fails, use a copy of the unique Excel worksheet as demonstrated .below

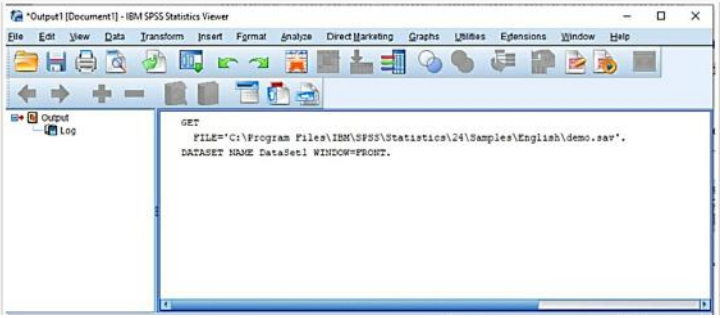


The Output/Viewer

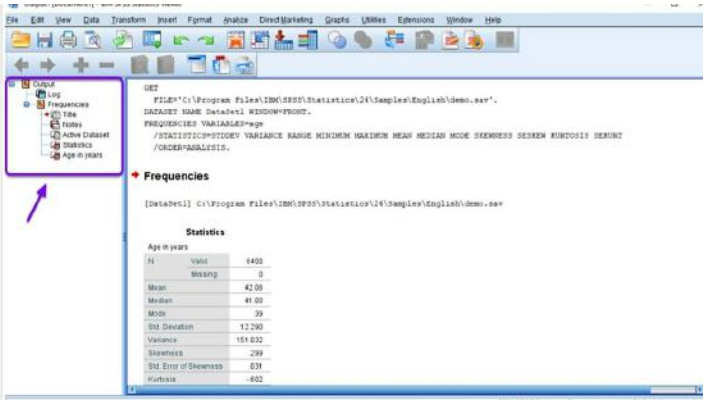
When you carry out any command in SPSS, the Output Viewer window opens mechanically and shows a log of the activities taken and the related output.

Mainly, the Output Viewer is where the outcomes of statistical calculation are expressed; however, any command conjured through the pull-down toolbar or syntax will be published to the Output Viewer. This includes viewing, shutting, or keeping a statistics file.

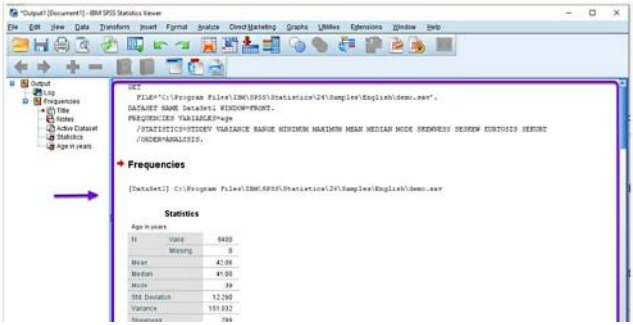
If an Output View window is not open when an instruction is performed, a new Output Viewer window will naturally be generated.



The Output Viewer window is split into segments. The left part includes a summary of the topic in the Output Viewer. This outline is specifically valuable if you have performed many SPSS instructions and want to find a specific segment of output effortlessly.

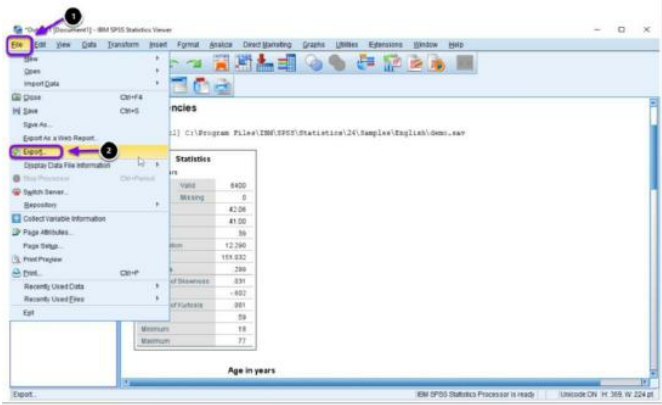


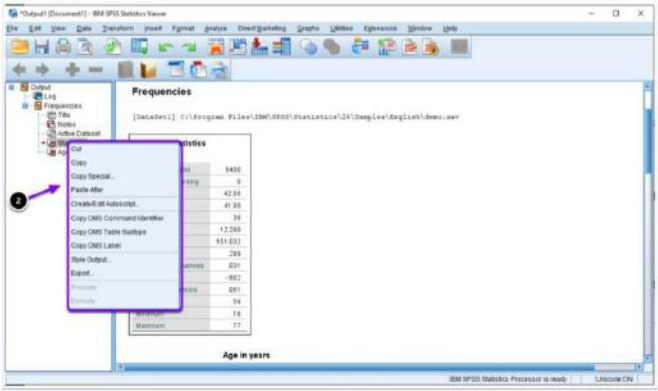
The exemplary segment includes the actual output. Tapping on an object in the left detail will cause it to bounce to that topic in the right piece. Objects chosen in the right part are denoted by a red arrow and a box drawn around the issue.



You can adjust the topics in the Output Viewer by choosing object(s) in the left or

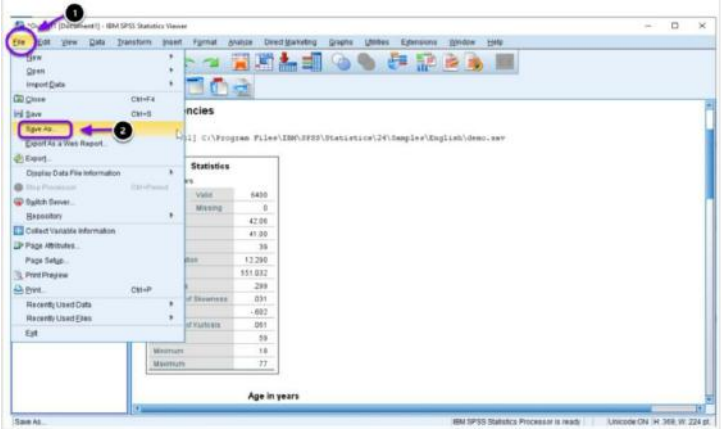
right segment, right-tapping the thing, and choosing the alternative you would love. To get rid of an object from the Output Viewer, tap on its title in the left segment or tap at the thing itself in the right part, then tap the Erase button to your keyboard.





An Output Viewer window may be stored as

a viewer document (*.spv), so you can evaluate it once more without re-do the exact instructions in SPSS. To keep an Output Viewer window, tap File and then Store As.



Select the area in which you would love your document stored using the pull-down arrow close to Look in: text topic box

Give a title to your document in the File name: text box.

Review other alternatives you may want.

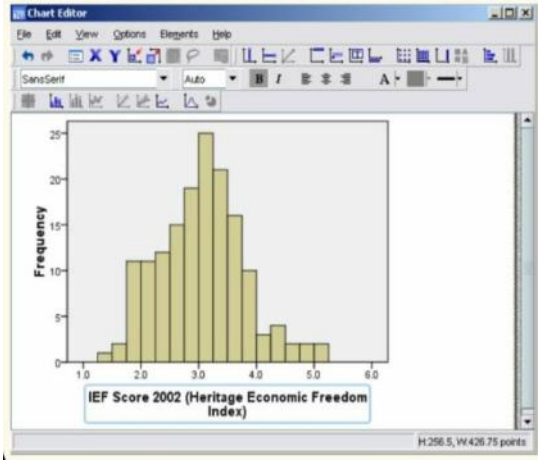
Tap Save.

Modify Chart Using

Chart Editor (APA Style)

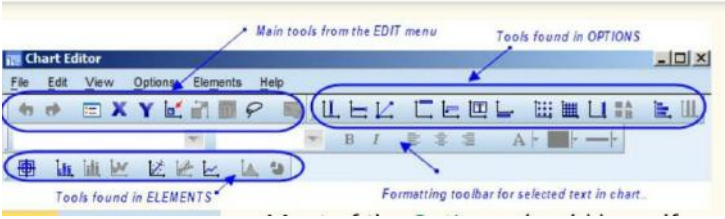
If you tap twice on a chart displayed in an SPSS Viewer window, you're starting the SPSS Chart Editor.

The new chart may be changed using the various tools provided. Retaining the same old File, you can Edit and View the toolbar(containing chart-specific objects. You may locate toolbars that allow you to carry out chart-specific tasks. Many are available from tool keys or converting the precise component in the Properties dialog. (Tapping twice on any chart component will cause the Properties dialog to open; choosing a specific feature at the chart indicates the alternatives accessible for it.

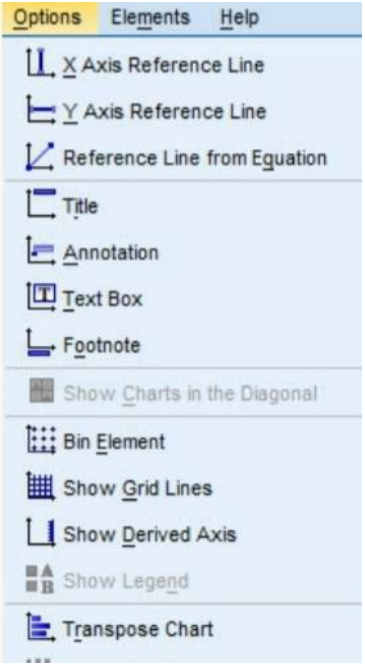


Tools and menus

The toolbars display the primary tools from numerous toolbars, except the configuration toolbar that conforms to alternatives you could locate in the properties layers for diverse textual components (titles, legends, annotations, etc.)



Hint: Tools and menu objects are grey, and the related alternative isn't always accessible for the new chart type.

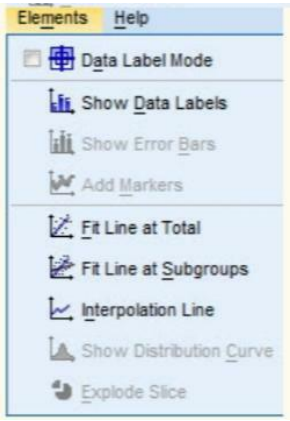


Horizontal and vertical reference lines may be positioned, both at specific user areas, at the average or the median of the related variables. In a scatterplot, you could upload more of your line primarily based totally on an equation you specify.

Titles, footnotes, and annotations are

positioned at fixed areas, while text boxes are freely positioned text.

The toolbar element gives more alternatives associated with details of specific chart categories, e.g., in scatterplots; you could tag observations, draw a fitted line, etc.



Display Data Labels shows case tags in scatterplots if a label variable has been distinctive.

Interpolation line connects character statistics points, e.g., on a scatter plot displaying a time series.

Hint:

1. Not every tool is essential and lively

for all kinds of charts.

2. Several alternatives may be put using the Properties window that springs up while choosing diverse elements in the chart. Hint: The tabs displayed in the Properties Window alternate every time you choose a specific part of the chart; in case you select, e.g., the scales, you may see a bill that helps you to alter ticks and additional axes appropriate objects.
3. Any time you rewrite a graph, it's displayed in a chart editor window; the chart in the Viewer window could be gray till you shut the chart editor. Hint: Numerous chart windows can be open simultaneously, and some of them are probably minimized or unseen at the back of various windows.

Modify Tables Using

Pivot Tables

Pivot tables

Most statistical techniques generate additional output frames (tables), known as Pivot tables. Frequency tables, cross-tabulations, degeneration output; however, the Case Processing overview are all pivot tables.

The pivot editor is a specialized editor for frames (output tables) generated by SPSS. Many specific tools you recognize from word processors or spreadsheets are accessible for modifying. They are informed about the particular shape of output developed by SPSS. Tables may be changed structurally and edited so that the table's frame stays orderly at any given time.

SPSS also generates easy frames like names, instruction logs, etc.; they may be edited with the Text editor

Starting the pivots editor

The output frames may be edited with the Pivot Editor opened by tapping twice click at

the complementary table or choosing the frame (single-tap on). Then, from the context toolbar, select Edit Content and then both In Viewer or In Separate Window.

Modify in Viewer

If you select In Viewer, the border of the frame modifications is displayed at the left. You may also note that initiating the pivot editor provides more objects to the Viewer toolbar, specifically a Pivot and Layout. Modifications you're making to the table are now seen in the Viewer window; you will most likely utilize this mode typically for small rewriting tasks, as easy as tapping on just outside the frame exists modifying method.

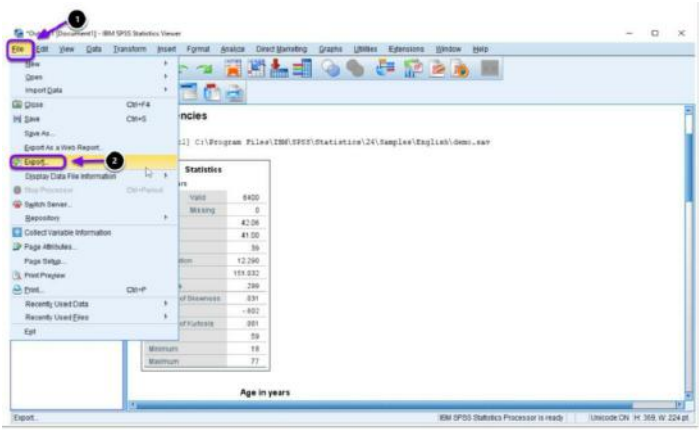
Modify in different Window

If you select to edit in a different window, a separate editor window will open up, showing the frame to edit. The table in the Viewer window will appear gray, as displayed in the backside image to the left to inform you that the structure is open in every other window. Modifications to the table

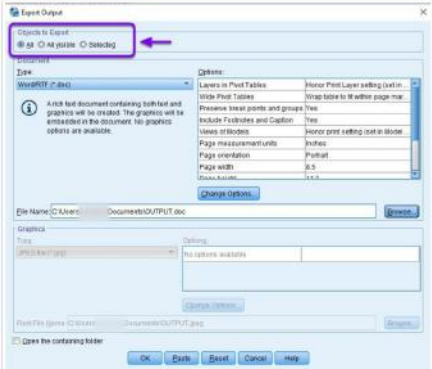
might be seen in the viewer window only once you Shut the Editor window.

Export Output

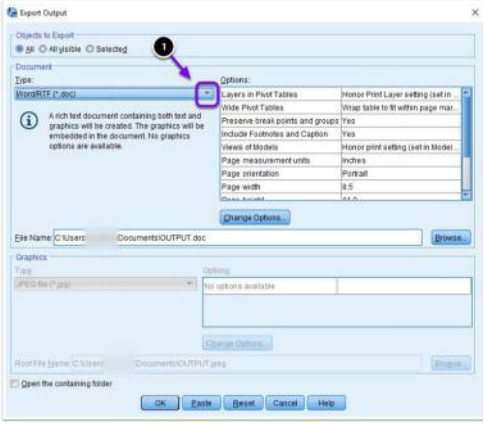
Preferably, you could export a few or all the topics in the Viewer window to a different file or image document by tapping File and then Export.

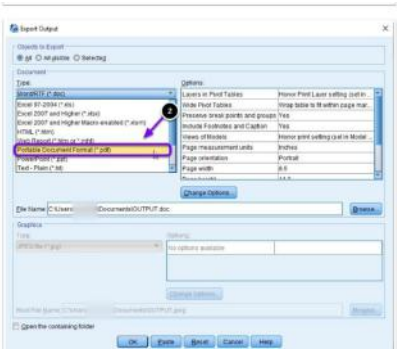


Choose which items you would love to export.

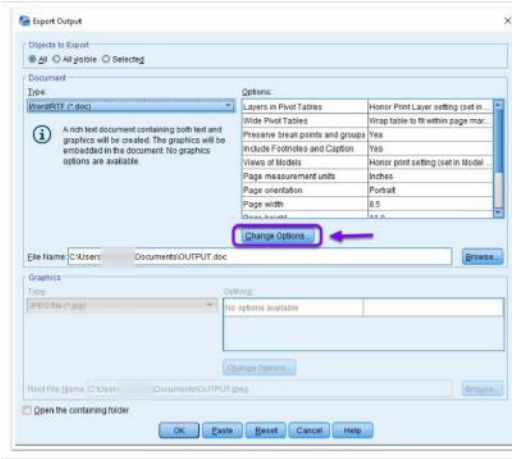


Select the kind of document you would love your items to be exported. Generally, you could export all topics as a PDF (*.pdf), a PowerPoint document (*.ppt), an Excel document (*.xls or *.xlsx), a Word document (*.document or *.docx), an HTML document (*.htm), or a text topic document (*.txt). Graphs may be stored as *.bmp, *.emf, *.eps, *.jpeg, *.png, or *.tif.

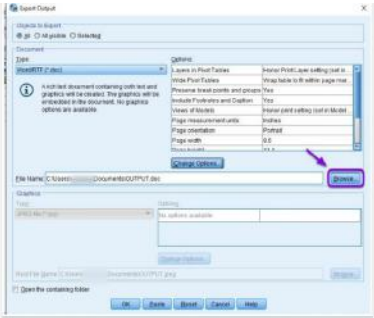




Discretionary: You can alter the alternatives of the exported item layout. Tap on Change Options. Select your options and tap Continue.



Tap on the Browse icon close to the File title: text topic box.

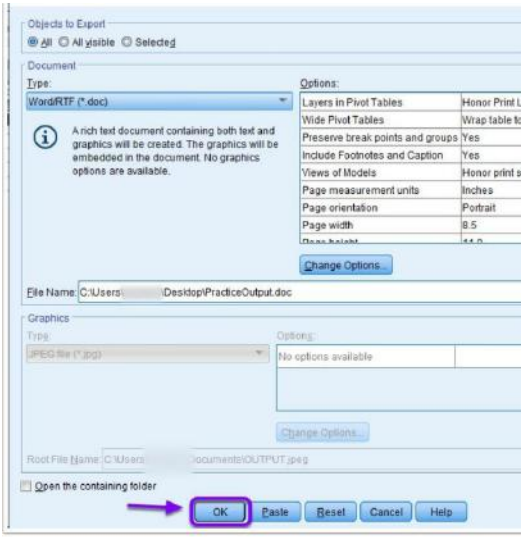


Choose an area to store the exported items by tapping the pull-down arrow close to the Look in text topic box

Give a title for the exported items in the File title: text topic box

Tap Save.

Lastly, tap on OK to export your items



Export Data

To export a given data into SPSS, you will need to follow the steps below

1. Build an export file.
2. To copy this file to SPSS, click on the 'SPSS Export file.'
3. Tap"export." The export file will be saved on your device in a folder

named:studyname_spss_export_datec

4. It is preferable to store this folder in an accessing but safe file area.
5. The file you have exported will be in a zip-file format containing numerous .dat and .sps files. To be able to import these files into SPSS, you need to unzip all the files (using "unzip" or "extract").
6. During the unzipping process, you will be asked to identify an area to save the files. This area is used later to import the files in SPSS, so ensure they are easy to locate.

From here, you can commence importing the data to SPSS using the steps below

1. Click on the zip file to open the syntax file with the name saved in your device earlier.
2. . This is seen in the fourth row in the format below. ppl/FILE='Your_study_name.dat.'

3. The screenshot below shows how SPSS displays the file when you open it.

```
1   SET UNICODE=ON.
2   GET DATA
3   /TYPE=TXT
4   /FILE='D:\Downloads\Export\BURROW_20160202.dat'
5   /DELCASE=LINE
6   /DELIMITERS=";"
7   /QUALIFIER=""
8   /ARRANGEMENT=DELIMITED
9   /FIRSTCASE=1
10  /IMPORTCASE=ALL
11  /VARIABLES=
12  record_id A12
13  institute_abbreviation A3
```

4. Ensure your document route is correct. Open the folder in which the .dat document is saved. In Windows, right-click on the document and tap 'Properties.' The document route is indexed close to 'Location.' In Mac, choose the document, then tap Command + i. The document route is indexed close to 'Where.'
5. If you use SPSS version 21 or higher, you need to eliminate the following

line: /QUALIFIER=’”’.

6. Choose the complete document with ctrl+a and perform this using the play lever in SPSS. The file will be imported into SPSS.
7. If, for some reason, the above steps don't work, it's wise to test the document route of the .dat document.

Exporting Survey Data Into SPSS

You can export your survey outcomes to utilize them in SPSS. SPSS exports are accessible for All Responses Data exports. Every Responses Data exports enable you to detect the comprehensive response batch for each respondent.

To export the outcomes of your survey, follow the steps below

1. Go to the icon that says Analyze Results segment of your survey.

2. Tap Save As at the top of the screen.
3. Tap Export document.
4. Choose the All Responses Data export category.
5. Select SPSS, choose your export choices, and tap Export.
6. Under EXPORTS in the left toolbar, tap the export to download the document to your device. Exports are saved here for 14 days.

Note: The export contains an SAV file. We suggest using SPSS 17.0.1 or a higher version to view the document.

Data Codebook

A codebook outlines the main statistics about the variables in a research program.

A codebook is a file containing data about every variable in your dataset, like:

The title assigned to the variable

What the variable indicates (label)

How the variable was calculated (e.g.,

nominal, ordinal, scale)

How the variable was undoubtedly documented in the raw information (i.e., numeric, string; the number of characters wide, the number of decimal settings it has)

For scale variables: The variable's units of dimension

For specific variables: If coded numerically, the numeric codes and what they imply.

Codebooks can also include documentation about how and when the statistics were developed. A suitable codebook enables you to convey your statistics studies to others definitely and concisely and ensures that the information is comprehended and inferred appropriately.

Many codebooks are developed manually; though, in SPSS, it is feasible to create a codebook from an occurring SPSS statistics document.

Developing a

Codebook from an SPSS Statistics Document

This codebook approach publishes most of the data located in the Variable View window. It provides the titles, tags, dimension levels, widths, configurations, and any selected omitting values tags for each variable in the dataset. It furthermore publishes a table with the value chosen labels for specific variables.

You can generate this easy codebook utilizing the point-and-tap toolbar or syntax.

Using the Toolbars

Open the SPSS data file.

Tap File > Show Data File Information > Working File.

The codebook will publish in the Output Viewer window.

Comprehensive Codebook

This codebook approach consists of all comparable data as the straightforward approach. However, it consists of alternatives for publishing summary information. Unlike the easy process, you can select which variables are contained in the codebook, and you can choose which variable characteristics are included in the summary. Also, unlike the straightforward approach, the summary statistics for every variable can be published in its table.

You can develop this comprehensive codebook using the dialogue window or syntax usage.

Note: This method was brought in SPSS version 17. If you're using an obsolete version of SPSS, this instruction is not accessible - it will not display in the toolbar, and performing the syntax will yield error statements.

Utilizing the Codebooks Dialog Window

Open the SPSS data document.

Tap Analyze > Reports > Codebook.

In the Variables tab: Include the variables you need in the codebook to the Codebook Variables box. To contain all variables, tap in the Variables box, tap Ctrl + A, then tap the arrow key.

In the Output tab: (Discretionary) Choose what variable and data document characteristics you need to be contained in the codebook:

Variable data: By default, contains Position, Tag Type, Layouts, Dimension level, Role, Value gs, Missing values, and Custom features.

File statistics: None contained by default.

Variable display arrangement default is

arranged similarly to how the variables are organized in the file. It can furthermore be arranged alphabetically, by document, or by dimension level.

The maximum range is restricted to two hundred types.

In the Statistics tab: (Discretionary) Select the information you need in the codebook, by default, scores, and percentages. It can be published for nominal and ordinal variables, median, standard deviation, and quartiles can be issued for scale variables.

When completed, tap OK.

Note: When listing the variable titles in the syntax, the selected dimension level ha. To be put in brackets after every variable title : [s] for scale, [n] for nominal, [o] for ordinal.

Using Syntax

```
CODEBOOK <variables-names-here>
  /VARINFO POSITION LABEL TYPE FORMAT MEASURE ROLE VALUELABELS MISSING
ATTRIBUTES
  /FILEINFO NAME CASECOUNT
  /OPTIONS VARORDER=VARLIST SORT=ASCENDING MAXCATS=200
  /STATISTICS COUNT PERCENT MEAN STDDEV QUARTILES.
```

Chapter Five
Inferential Statistics

T-Tests

1. One-Sample T-Test

The One-Sample t-Test correlates a sample's average to a hypothesized value for the populace average to deduce whether or not the 2 methods are considerably unique.

The One-Sample t-Test evaluates whether or not the average of a populace is statistically unique from a recognized or hypothesized value. The One-Sample t-Test is a parametric check.

This test is also recognized as:

Single Sample t-Test

The variable utilized in this test is called:

Test variable

In a One-Sample t-Test, the test variable's

average is differentiated from a "test value", which is a recognized or hypothesized value of the suggestions in the populace. Test values may also come from a literature study, a reliable studies institution, lawful requirements, or enterprise standards. For instance:

A specific plant's machines are speculated to fill bottles with 0.15 Liters of the commodity. A plant supervisor desires to check a random sample of bottles to make sure that the machines aren't under- or over-filling the bottles.

An inspector needs to test if samples taken from divisions in a condo surpass the clearance level.

General Uses

The One-Sample t-Test is typically used to check the following:

The statistical distinction between an average and a recognized or hypothesized value of the average in the populace.

The statistical discrepancy between a

modification mark and zero.

This technique includes developing an alternate rating from variables, and then distinguishing the average fluctuation rating to zero, to infer whether or not any alternate existed among the 2-time facts for the unique criteria. If the average modification rating isn't appreciably distinct from zero, no giant alternate befell.

Hint: The One-Sample t-Test can only distinguish a single sample average from a particular constant. It cannot examine sample averages among or other groups.

Data Criteria

1. Your statistics should satisfy the following criteria: Consecutive test variables.
2. Records on the test variable are independent (i.e., independence of observations)
3. There is no association among rankings on the test variable
4. Breach of this hypothesis will

generate a faulty value.

5. Spontaneous sample of statistics from the populace normal measurement of the sample and populace at the test variable
6. Non-regular populace measurements, especially the ones which might be thick-tailed or closely skewed, considerably decrease the energy of the test.
7. Among mild or big samples, an infringement of normality may also nevertheless yield correct p values

Uniformity of variances (i.e., variances about the same in each sample and populace)

No outliers

Theories

The null hypothesis (H_0) and (two-tailed) alternative hypothesis (H_1) of the one sample T test can be expressed as:

H_0: $\mu = \mu_0$ ("the population mean is equal to the [proposed] population mean")
H_1: $\mu \neq \mu_0$ ("the population mean is not equal to the [proposed] population mean")

where μ is the "true" population mean and μ_0 is the proposed value of the population mean.

The Statistic Check

The statistic Check for a One-Sample t-Test is denoted by t. It is evaluated using the formula below:

$$t = \frac{\bar{x} - \mu_0}{s_{\bar{x}}}$$

where

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

where

μ_0 = The *test value* -- the proposed constant for the population mean
$\bar{x}$ = Sample mean
n = Sample size (i.e., number of observations
s = Sample standard deviation
$s_{\bar{x}}$ = Estimated standard error of the mean (s/sqrt(n))

The calculated t value is then distinguished from the critical t value from the t distribution table with levels of freedom df = n - 1 and selected self-reliance level. If the calculated t value > significant t value, then we dismiss the null hypothesis.

Data Set-Up

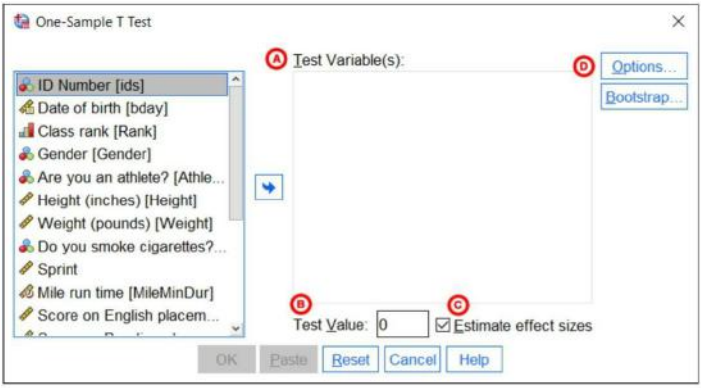
Your facts must consist of one constant, numeric variable (illustrated in a column) that will be utilized in the evaluation. The variable's dimension level must be described as Scale in the Variable View window.

Perform a One-Sample T-Test

To perform a One-Sample TTest in SPSS, tap Analyze > Compare Means > One-Sample T-Test.

The One-Sample T-Test window opens where you will define the variables to be used in the evaluation. All of the variables on your dataset are seen in the listing on the left side. Shift variables to the Test Variable(s) region

by choosing them in the listing and tapping the arrow icon.



A Test Variable(s): The variable whose average could be correlated to the hypothesized populace average. You may also perform many One-Sample t-Tests concurrently by choosing multiple test variables. Every variable will be correlated to the same Test Value.

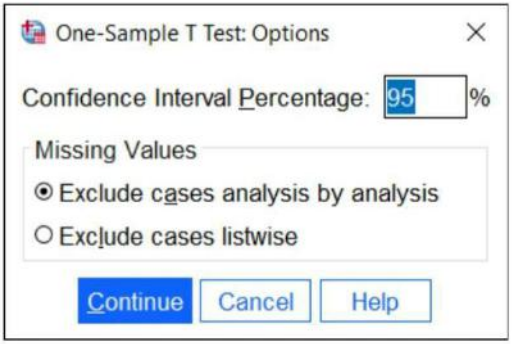
B Test Value: The hypothesized populace average against which your test variable(s) will be as correlated.

C Estimate effect sizes: discretionary. If tested, will publish effect length statistics --

namely, Cohen's d -- for the check(s).

D Options: tapping Options will start a window in which you could define the Confidence Interval Percentage and the way the evaluation will deal with Missing Values.

Tap Continue while you are done creating specifications



Tap OK to perform the t-Test.

2. Paired Samples T-TEST

Paired t assessments are utilized to check if the method of paired computations, including that of pretest/posttest marks, are considerably specific.

The Paired Samples t-Test correlates the method of computations taken from the same individual, object, or associated devices. These "paired" computations can constitute topics like:

1. A computation is taken at specific instances.
2. A computation took place under specific situations.
3. Computations took from halves or aspects of a patient or experimental unit.

The reason for the test is to deduce whether or not there is statistical proof that the average distinction among paired observations is notably distinct from zero. The Paired Samples t-Test is a parametric check.

This test is also recognized as:

1. Dependent t-Test
2. Paired t-Test
3. Repeated Measures t-Test

The variable used in this test is called :

The dependent variable, or test variable (constant), is estimated at specific instances or for 2 associated situations or units.

General Uses

The Paired Samples t-Test is normally used to check the following:

1. The statistical disparity between time points
2. The statistical disparity between situations
3. The statistical disparity between quantities
4. The statistical disparity between a matched pair

Hint: The Paired Samples t-Test isn't always suitable for calculations about the following: 1) unpaired information;

2) Differentiations between multiple units/groups;

3) continuous last outcomes that aren't always typically dispensed, and

4) ordinal/ranked last outcomes.

Data Criteria

Your information should satisfy the following criteria:

1. Dependent variable this is continuous (i.e., period or rate level)Hint: The paired contents should be documented in different variables.
2. Corresponding samples/groups (i.e., based observations)
3. The topics in every pattern, or institution, are the same. This means that the topics in the first institution also are in the 2d group.
4. An arbitrary sample of information from the population normal distribution (approximately) of the distinction among the paired values.
5. No outliers in the disparity between the 2 associated groups.

Hint: When checking out hypotheses associated with normalcy and outliers, you

should use a variable that depicts the disparity between the paired values - not the unique variables themselves.

Hint: When one or more of the hypotheses for the Paired Samples t-Test aren't met, you may need to perform the nonparametric Wilcoxon Signed-Ranks Test.

Assumptions

The assumptions may be conveyed in specific approaches that imply the same concept and are mathematically equivalent:

H_0: $\mu_1 = \mu_2$ ("the paired population means are equal")
H_1: $\mu_1 \neq \mu_2$ ("the paired population means are not equal")

OR

H_0: $\mu_1 - \mu_2 = 0$ ("the difference between the paired population means is equal to 0")
H_1: $\mu_1 - \mu_2 \neq 0$ ("the difference between the paired population means is not 0")

μ_1 is the population mean of variable 1,
and
μ_2 is the population mean of variable 2.

The test statistic for the Paired Samples t-Test, signified as t, interprets the same formula as the one-sample t-test.

$$t = \frac{\bar{x}_{\text{diff}} - 0}{s_{\bar{x}}}$$

where

$$s_{\bar{x}} = \frac{s_{\text{diff}}}{\sqrt{n}}$$

where

$\bar{x}_{\text{diff}}$ = Sample mean of the differences
n = Sample size (i.e., number of observations)
s_{diff} = Sample standard deviation of the differences
$s_{\bar{x}}$ = Estimated standard error of the mean (s/sqrt(n))

The calculated t value is then compared to the critical t value with df = n - 1 from the t distribution table for a chosen confidence level. If the calculated t value is greater than the critical t value, then we reject the null hypothesis (and conclude that the means are significantly different).

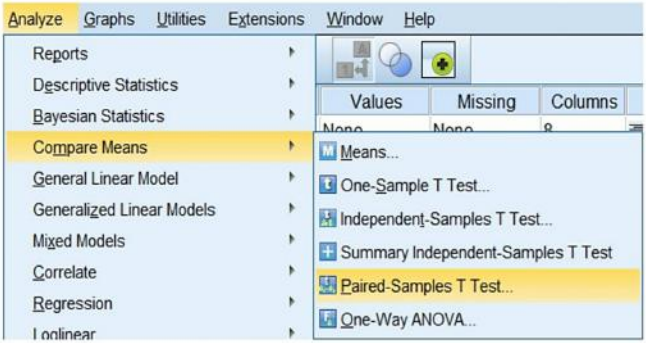
Setting Up Your Data

Your records must contain consecutive numeric variables (denoted in columns) that will be utilized in the evaluation. The

variables must consist of the paired variables for every subject (row). If your records are organized otherwise (e.g., cases signify repeated units/subjects), simply restructure the records to indicate this layout.

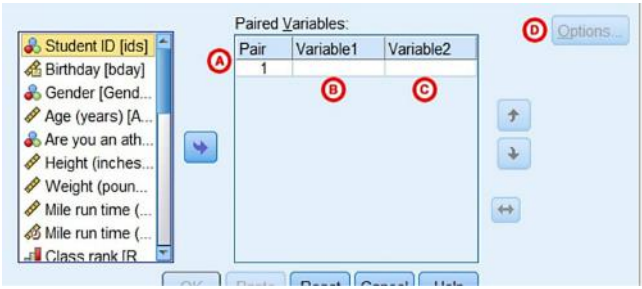
Perform a Paired Samples t-Test

To perform a Paired Samples t-Test in SPSS, tap on Analyze > Compare Means > Paired-Samples T-Test.



The Paired-Samples T-Test window opens where you'll define the variables to be utilized in the evaluation. All variables in your dataset are seen in the listing on the left

side. Shift variables to the right by choosing them in the listing and tapping the blue arrow icons. You will define the paired variables in the Paired Variables region.



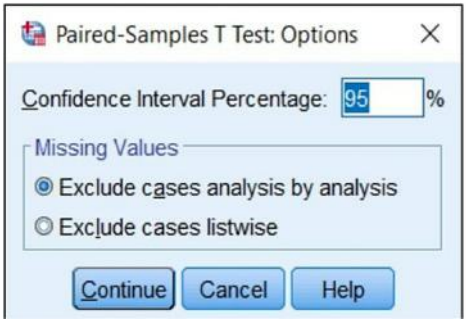
Paired-Samples T-Test window In SPSS

A Pair: The "Pair" column signifies the variety of Paired Samples t-Tests to run. You may select to perform numerous Paired Samples t-Tests concurrently by selecting more than one unit of matched variables. Each new pair will emerge on a new line.

B Variable1: The first variable, denoting the main group of matched values. Shift the variable that denotes the main group to the right where it will likely be documented below the "Variable1" column.

C Variable2: The 2nd variable, denoting the second group of matched values. Shift the variable that denotes the second group to the right where it will likely be documented below the “Variable2” column.

D Options: Tapping Options will open a window where you can define the Confidence Interval Percentage and the way the evaluation will deal with Missing Values (i.e., Exclude instances evaluation by evaluation or Exclude instances listwise). Tap Continue when you are done creating the specifications.



Assigning the confidence interval percentage does not have any effect on the analysis of the p-value.

If you are only performing one paired sample t-test, both assigned"missing values" will

generate the same outcomes. There will only be disparities if you are performing multiple paired samples t-tests.

3. **The Independent Samples t-Test** This Correlates sample averages to deduce whether or not the populace averages are substantially different.

The Independent Samples t-Test correlates the averages of independent groups to specify whether or not there may be statistical proof that the related populace means are substantially different. The Independent Samples t-Test is a parametric check.

This test is recognized as:

a. Independent t-Test
b. Independent Measures t-Test
c. Independent Two-sample t-Test
d. Learner t-Test
e. Two-Sample t-Test
f. Uncorrelated Scores t-Test
g. Unpaired t-Test
h. Unrelated t-Test

The variables utilized in this check are called:

The dependent variable, or test variable,

Independent variable, or grouping variable

General Uses

The Independent Samples t-Test is typically utilized to test the following:

1. Statistical variations among the averages of groups
2. Statistical variations among the averages of interventions
3. Statistical variations among the averages of alternate scores.

Data Criteria

Your data ought to satisfy the following criteria:

1. Consecutive dependent variable (i.e., period or percentage level)
2. Independent variable that is categorical.
3. Cases that have values on both the

dependent and independent variables

4. Independent samples/groups. There is no association between the topics in every sample. This implies that:

a. Subjects in the initial group can't also be in the second group
b. No subject in both groups can affect subjects in the different group
c. No group can affect the alternative group.
d. Infringement of this hypothesis will result in an erroneous p-value.

5. Spontaneous sample of facts from the populace.

6. Normal distribution of the dependent variable for every group.

a. Non-regular populace distributions, especially the ones which are thick-tailed or closely skewed, notably lessen the energy of the test.

b. Among mild or big samples, an infringement of normalcy may also still result

in correct p values.

7. Uniformity of variances (i.e., variances is roughly the same throughout the groups)

When this assumption is infringed and the sample magnitudes for every group vary, the p-value isn't reliable.

Nonetheless, the Independent Samples t-Test output also consists of an estimated t statistic that isn't based on speculating the same populace variances. This alternative statistic, known as the Welch t-Test statistic1, can be used when the same variances among populations can't be deduced. The Welch t-Test is called an Unequal Variance t-Test or Separate Variances t-Test.

8. No outlines.

Hint: When one or more of the hypotheses for the Independent Samples t-Test aren't fulfilled, you may need to perform the nonparametric Mann-Whitney U test instead.

Experimenters frequently observe numerous guidelines of thumb:

1. Each group ought to have a minimum of 6 subjects, preferably more. Hypotheses for the populace might be more insubstantial with extremely limited subjects.
2. Symmetrical design (i.e., the same quantity of subjects in every group) is suitable. Extremely unstable designs improve the chance that infringing any of the criteria will endanger the certainty of the Independent Samples t-Test.

The Assumptions

The null hypothesis (H0) and alternative hypothesis (H1) of the Independent Samples t-Test can be formulated in two different but the same ways:

H_0: $\mu_1 = \mu_2$ ("the two population means are equal")
H_1: $\mu_1 \neq \mu_2$ ("the two population means are not equal")

OR

H_0: $\mu_1 - \mu_2 = 0$ ("the difference between the two population means is equal to 0")
H_1: $\mu_1 - \mu_2 \neq 0$ ("the difference between the two population means is not 0")

where μ_1 and μ_2 are the population means for group 1 and group 2, respectively. Notice that the second set of hypotheses can be derived from the first set by simply subtracting μ_2 from both sides of the equation.

Setting Up Data

Your statistics have to contain variables (denoted in columns) that will be utilized in the calculation. The independent variable has to be specific and contain precise groups.

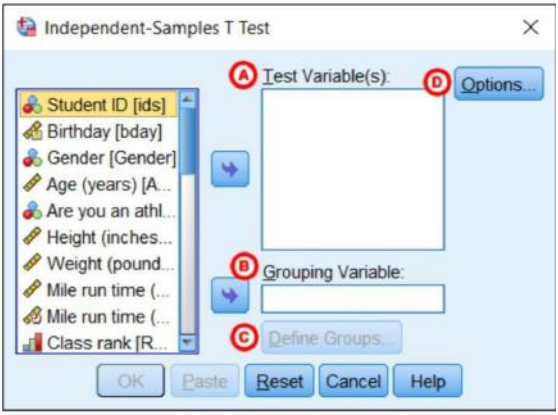
The variety of rows in the dataset has to conform to the variety of subjects in the research. Each row of the dataset has to portray a different subject, individual, or unit, and all the computations taken on that individual or unit have to appear in that row.

Perform an Independent Samples t-Test

To perform an Independent Samples t-Test in

SPSS, tap Analyze > Compare Means > Independent-Samples T-Test.

The Independent-Samples T-Test window opens where you will define the variables to be utilized in the analysis. All of the variables to your dataset seem withinside the listing on the left side. Shift variables to the right by choosing them in the listing and tapping the blue arrow icons. You can shift a variable(s) to both regions: Grouping Variable or Test Variable(s).



A Test Variable(s): The dependent variable(s). This is the continuous variable whose averages will be as correlated among the 2 groups. You may perform more than

one t-tests concurrently by choosing multiple test variables.

B Grouping Variable: The independent variable. The classes (or groups) of the independent variable will characterize which samples will be compared in the t-test. The grouping variable need to have the least categories (groups); it could have greater categories., however, a t-test can only distinguish groups so that you will need to identify which groups to examine. You also can use a continuous variable by defining a cut factor to generate groups (i.e., values at or above the cut factor and values beneath the cut factor).

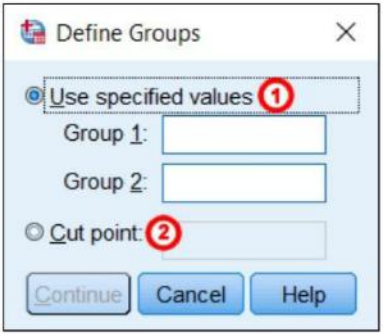
C Define Groups: Tap Define Groups to define the category indicators (groups) to apply in the t-test. If the key isn't active, ensure that you have already shifted your independent variable to the right in the Grouping Variable field. You ought to outline the kinds of your grouping variable earlier than you could run the Independent Samples t-Test procedure.

D Options: The Options segment is where you could put your preferred confidence stage for the confidence interval for the average disparity, and determine how SPSS has to manage missing values.

When completed, tap OK to perform the Independent Samples t-Test, or tap Paste to have the syntax similar to your unique settings written to an open syntax window. (If you do not have a syntax window open, a new window will open for you.)

Define Group

Tapping the Define Groups button (C) opens the Define Groups window:



1 Use unique values: If your grouping variable is specific, choose Use specified

values. Enter the values for the types you want to examine in the Group 1 and Group 2 areas. If your classes are numerically coded, you will input the numeric codes. If your group variable is a string, you will input the precise textual content strings denoting the 2 classes. If your grouping variable has more than two classes (e.g., takes on values of 1, 2, 3, 4), you could determine the types to be compared.

2 Cut factor: If your grouping variable is numeric and continuous, you can select a cut factor for separating the variable. This will separate the cases into classes based on the cute factor. Particularly, for a given cut factor x, the new classes may be:

Group 1: All instances where grouping variable > x

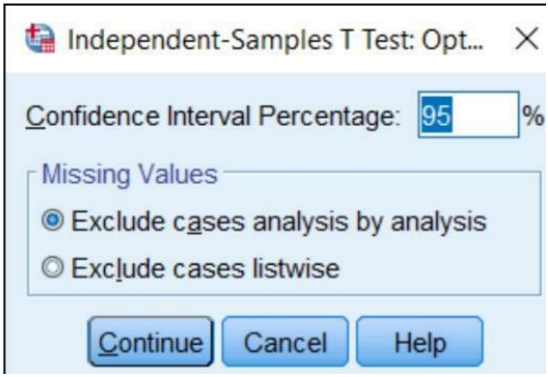
Group 2: All instances in which grouping variable < x

This means that the case, where the grouping variable is identical to the cut factor itself, will be contained in the "more than or identical to" class. (If you need your cut

factor to be contained in a "less than or identical to" group, then you will want to apply to Recode into Different Variables or utilize DO IF syntax to generate this grouping variable yourself.)

Alternative.

Tapping the Options icon(D) opens the Options window:



The Independent Samples T-Test Options window enables you to adjust the confidence interval percent and selects between listwise or 'evaluation by evaluating' (pairwise) missing records procedure.

The Confidence Interval Percent box enables you to stipulate the confidence level for a

confidence interval. Hint: This setting does NOT influence the test statistic or p-value or preferred error; it only influences the computed top and decrease bounds of the confidence interval. You can input any value between 1 and ninety-nine on this box (even though in practice, it is sensible to input only numbers between ninety and ninety-nine).

The Missing Values area enables you to select if cases need to be eliminated "evaluation by evaluation" (i.e. pairwise delete) or eliminated listwise. This setting isn't applicable when you have only one particular dependent variable; it only matters if you are inputting a couple of dependents (continuous numeric) variables. In that case, dismissing "evaluation by evaluation" will utilize all nonmissing values for a given variable. If you dismiss "listwise", it'll only employ the cases with nonmissing values for all the variables inputted. Depending on the number of missing records you have, listwise deletion should significantly decrease your sample size.

Levenger's Equal Variance

Remember that the Independent Samples t-Test needs the hypothesis of uniformity of variance -- i.e., both groups have the identical variance. SPSS readily consists of a test for the uniformity of variance, referred to as Levene's Test, each time you perform an independent sample t-test.

The assumptions for Levene's test are:

H_0: $\sigma_1^2 - \sigma_2^2 = 0$ ("the population variances of group 1 and 2 are equal")

H_1: $\sigma_1^2 - \sigma_2^2 \neq 0$ ("the population variances of group 1 and 2 are not equal")

This means that if we dismiss the null assumption of Levene's Test, it shows that the variances of the 2 groups aren't the same; i.e., that the uniformity of variances hypothesis is infringed.

The output in the Independent Samples Test table consists of rows: Equal variances

inferred and Equal variances not inferred. If Levene’s test shows that the variances are the same throughout the 2 groups (i.e., the p-value will be big), you may depend on the main row of output, Equal variances inferred, when you study the outcomes for the real Independent Samples t-Test (under the title t-test for Equality of averages). If Levene’s test indicates that the variances aren't the same throughout the 2 groups (i.e., the p-value will be small), you will have to depend on the second one row of output, Equal variances not inferred, while you study the effects of the Independent Samples t-Test (under the heading t-test for Equality of Averages).

The distinction between these rows of output lies in the manner the independent samples t-test calculation is evaluated. When the same variances are inferred, the calculation makes use of pooled variances; when the same variances can't be assumed, the calculation makes use of un-pooled variances and a revision to the levels of freedom.

Test Statistics

The test statistic for an Independent Samples t-Test is signified by t. There are simply forms of the test statistic for this test, relying on whether or not the same variances are inferred. SPSS generates both forms of the test, so each form of the test is defined here. Hint: The null and alternative assumptions are the same for each form of the test calculation.

Equal Variances Inferred

When the 2 independent samples are inferred to be brought from populations with the same populace variances (i.e., σ12 = σ22), the test calculation t is evaluated as:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

with

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Where

$\bar{x}_1$ = Mean of first sample

$\bar{x}_2$ = Mean of second sample

n_1 = Sample size (i.e., number of observations) of first sample

n_2 = Sample size (i.e., number of observations) of second sample

s_1 = Standard deviation of first sample

s_2 = Standard deviation of second sampl

s_p = Pooled standard deviation

The calculated t value is then compared to the critical t value from the t distribution table witl degrees of freedom $df = n_1 + n_2 - 2$ and chosen confidence level. If the calculated t value is greater than the critical t value, then we reject null hypothesis.

EQUAL VARIANCES NOT ASSUMED

When the two independent samples are inferred to be pulled from populations with different variances (i.e., σ12 ≠ σ22), the test statistics t is evaluated as:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where

$\bar{x}_1$ = Mean of first sample
$\bar{x}_2$ = Mean of second sample
n_1 = Sample size (i.e., number of observations) of first sample
n_2 = Sample size (i.e., number of observations) of second sample
s_1 = Standard deviation of first sample
s_2 = Standard deviation of second sampl

The calculated t value is then compared to the critical t value from the t distribution table witl degrees of freedom

One-Way ANOVA

This Correlates sample averages to deduce whether or not the populace averages are substantially different.

The Independent Samples t-Test correlates the averages of independent groups to specify whether or not there may be statistical proof that the related populace means are substantially different. The One-Way-ANOVA Samples t-Test is a parametric check

.This test is also called:

1. One-Factor ANOVA
2. One-Way Analysis of Variance

Between Subjects ANOVA

The variables utilized in this test are called :

Dependent variable

Independent variable.

This variable separates cases into two or more jointly limited phases, or groups.

General Uses

The One-Way ANOVA is always utilized to evaluate information from the following kinds of studies:

Field studies

Investigations

Quasi-experiments

The One-Way ANOVA is normally utilized to test the following:

1. Statistical variations among the

averages of groups
2. Statistical variations among the averages of interventions
3. Statistical variations among the averages of alternate scores.

Hint: If the assemblage variable has only two groups, then the outcomes of a one-way ANOVA and the independent samples t-test will be the same. If you perform both an independent samples t-test and a one-way ANOVA in this circumstance, you should be able to corroborate that t^2=F.

Data Criteria

Your data ought to satisfy the following criteria:

5. Continuous dependent variable (i.e., period or percentage level)
6. Independent variable that is categorical (i.e., or more groups)
7. Cases that have values on both the dependent and independent variables
8. Independent samples/groups (i.e.,

independence of observations). There is no association between the topics in every sample. This implies that:

e. Subjects in the initial group can't also be in the second group
f. No subject in both groups can affect subjects in the different group
g. No group can affect the alternative group.
h. Infringement of this hypothesis will result in an erroneous p-value.

5. Spontaneous sample of facts from the populace.

6. Normal distribution (approximately) of the dependent variable for every group.

a. Non-regular populace distributions, especially the ones which are thick-tailed or closely skewed, notably lessen the energy of the test.

b. Among mild or big samples, an infringement of normalcy may also still result in correct p values.

7. Uniformity of variances (i.e., variances is roughly the same throughout the groups)

When this assumption is infringed and the sample magnitudes for every group vary, the p-value isn't reliable.

Nonetheless, the Independent Samples t-Test output also consists of an estimated t statistic that isn't based on speculating the same populace variances. This alternative statistic, known as the Welch t-Test statistic1, can be used when the same variances among populations can't be deduced. The Welch t-Test is called an Unequal Variance t-Test or Separate Variances t-Test.

8. No outlines.

Experimenters frequently observe numerous guidelines of thumb:

1. Each group ought to have a minimum of 6 subjects, preferably more. Hypotheses for the populace might be more insubstantial with extremely limited subjects.

2. Symmetrical design (i.e., the same quantity of subjects in every group) is suitable. Extremely unstable designs improve the chance that infringing any of the criteria will endanger the certainty of the ANOVA F-Test.

Assumptions

The null and alternative assumptions of one-way are mathematically written as follows:

H_0: $\mu_1 = \mu_2 = \mu_3 = \ldots = \mu_k$ ("all k population means are equal")

H_1: At least one μ_i different ("at least one of the k population means is not equal to the others")

where

- μ_i is the population mean of the ith group ($i = 1, 2, \ldots, k$)

Hint: The One-Way ANOVA is regarded as an omnibus (Latin for “all”) test because the F test demonstrates whether the prototype is substantial overall—i.e., whether or not there are any substantial disparities in the averages between any of the groups.

Setting Up Data

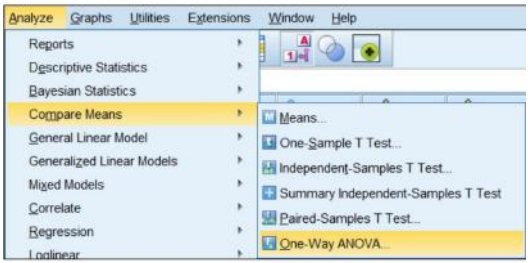
Your statistics have to contain variables (denoted in columns) that will be utilized in the calculation. The independent variable has to be specific and contain precise groups. (Hint: SPSS prohibits specific signs to numeric or short string values only.)

Perform a One-Way ANOVA

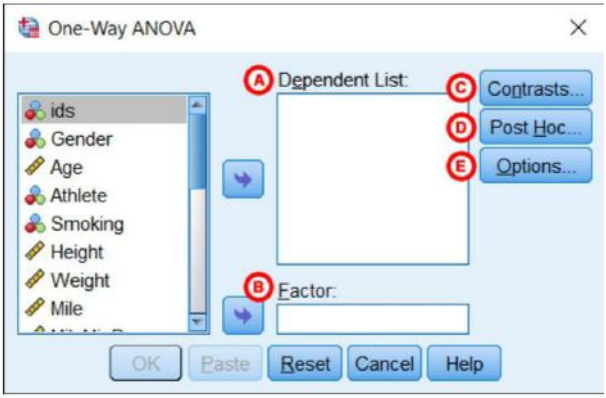
The following steps examine SPSS's devoted One-Way ANOVA technique. Nonetheless, since the One-Way ANOVA is also a component of the General Linear Model (GLM) class of statistical assessments, it may likewise be performed through the Univariate GLM technique ("univariate" refers to 1 dependent variable). This latter technique can be useful if your evaluation is going past the easy One-Way ANOVA and includes more than one independent variable, constant and spontaneous factors, and/or weighting variables and covariates (e.g., One-Way

ANOVA). We proceed by clarifying the way to perform a One-Way ANOVA using SPSS's devoted technique.

To perform a One-Way ANOVA in SPSS, tap Analyze > Compare Means > One-Way ANOVA.



The One-Way ANOVA window opens, where you'll define the variables to be used in the evaluation. Every variable in your dataset emerges in the listing on the left side. Shift variables to the right by choosing them in the listing and tapping the blue arrow icon. You can shift a variable(s) to either of the regions: Dependent List or Factor.



See the screenshot above.

A Dependent List: The dependent variable(s). This is the variable whose averages can be in comparison among the samples (groups). You may perform more than one average comparison concurrently by choosing multiple dependent variables.

B Factor: The independent variable. The categories (or groups) of the independent variable will outline which samples can be compared. The independent variable should have at least categories (groups), however normally has 3 or more groups when utilized in a One-Way ANOVA.

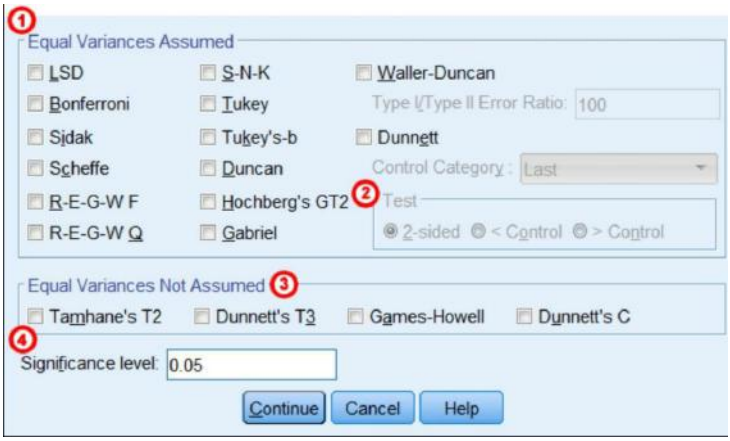
C Contrasts: (Discretionary) define

contrasts, or planned comparisons, to be performed after the general ANOVA check.

See the screenshot above.

When the preliminary F test shows that large variations occur among group averages, contrasts are beneficial for figuring out which precise averages are considerably unique if you have precise assumptions that you want to check. Contrasts are determined before evaluating the data (i.e., a priori). Contrasts disintegrate the variance into factor parts. They may include using weights, non-orthogonal comparisons, trendy contrasts, and polynomial contrasts (trend evaluation).

D Post Hoc: (Discretionary) Request post hoc (also called more than one comparison) assessments. Distinct post hoc assessments may be chosen by checking the related boxes.



See the screenshot above.

1 Equal Variance Assumed: Multiple comparisons alternatives that count on uniformity of variance (every group has the same variance). For comprehensive records about the precise evaluation procedures, tap the Help button on this window.

2 Test: By default, a 2-sided assumption check is chosen. Alternatively, a directional, one-sided assumption check may be documented if you select to apply a Dunnett

submit hoc check. Tap the box close to Dunnett and then determine whether or not the Control Category is the Last or main group, numerically, of your grouping variable. In the Assessment region, tap both < Control> Control. The one-tailed alternatives require that you determine whether or not you anticipate that the average for the required control group can be less than (> Control) or more than (< Control) any other group.

three Equal Variances Not inferred: Multiple comparisons alternatives that do not assume the same variances. For comprehensive data about the precise evaluation procedures, tap the Help button on this window.

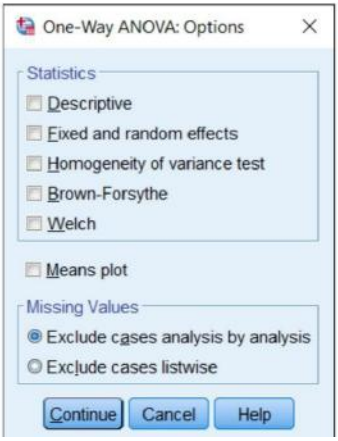
four Significance level: The preferred cutoff for statistical importance. By default, the importance is about 0.05.

When the preliminary F test shows that large variations exist among group averages, post hoc assessments are beneficial for figuring out which precise averages are considerably unique while you do not have precise

assumptions which you want to test. Post hoc assessments evaluate every pair of averages (like t-tests), however, unlike t-tests, they rectify the importance estimate to account for the multiple comparisons.

E Options: tapping Options will generate a window where you could determine which Statistics to include in the output (Descriptive, Fixed and random effects, Uniformity of variance test, Brown-Forsythe, Welch), whether or not to include an Averages plot, and the way the evaluation will deal with Missing Values.

Tap Continue when you are done creating specifications.



Tap OK to perform the One-Way ANOVA.

Two-Way ANOVA

A Two Way ANOVA in SPSS has independent variables. You might use a way ANOVA if you have one distribution (scale) variable and nominal variables. In the hypothetical instance that follows, we're analyzing how the brand of cleanser and water temperature (each nominal variable) affect cleanliness level (a scale variable).

Performing the Test

In this experiment, manufacturers of cleansers have been compared to see how well they can keep kitchen pans clean. The manufacturers have been examined in distinctive water temperatures: warm and cold. Brand of cleanser is the independent variable and level of cleanliness is the dependent variable. Water cleansers may have an impact on how well the manufacturers clean, so that could be a 2nd primary impact under deliberation. The interaction of Water Temperature and Brand

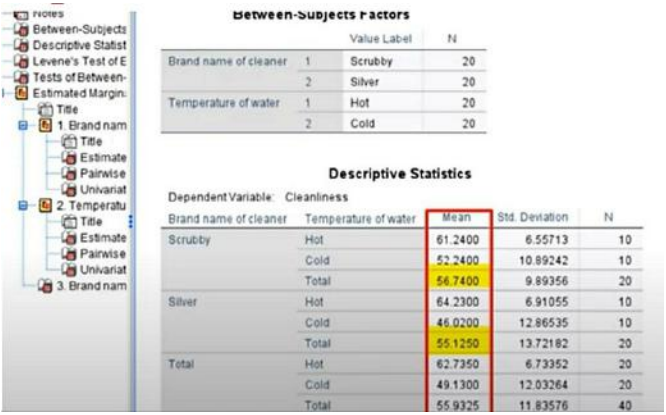
may have an interplay impact and so is included in the calculation.

1. Tap Analyze, choose General Linear Model, after which, tap Univariate.
2. TapYour dependent variable (in this instance, that's Cleanliness), after which tap the arrow next to the "Dependent Variable box".
3. Tap your primary independent variable—the primary focus of your investigation (for this instance, that's Brand of Cleaner, after which tap the arrow for the Fixed Factor(s): box.
4. Tap your interplay variable (Water temperature on this instance), after which tap the arrow for Fixed Factor(s).
5. Tap the EM Means box.
6. Tap your independent variable ("Brands") in the Univariate Estimated Marginal Means window. Press down Ctrl on your keyboard after which tap Water and

Brand*Water. Tap the arrow to shift all 3 to the Display Means for the box.

7. Tap Compare primary outcomes.
8. Tap Continue.
9. Tap Options.
10. Tap Descriptive information and Uniformity tests. You can alternate the alpha stage here, however, in this instance, I'm going to leave it on the 5% stage.
11. Tap Continue after which tap OK.

Examining the Outcomes in the Output Viewer window



Between-Subjects Factors

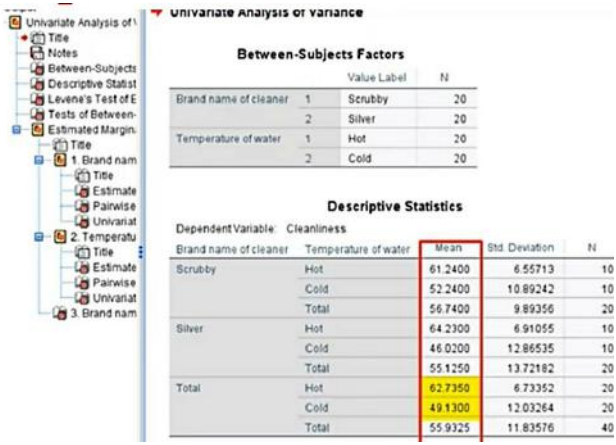
		Value Label	N
Brand name of cleaner	1	Scrubby	20
	2	Silver	20
Temperature of water	1	Hot	20
	2	Cold	20

Descriptive Statistics

Dependent Variable: Cleanliness

Brand name of cleaner	Temperature of water	Mean	Std. Deviation	N
Scrubby	Hot	61.2400	6.55713	10
	Cold	52.2400	10.89242	10
	Total	56.7400	9.89356	20
Silver	Hot	64.2300	6.91055	10
	Cold	46.0200	12.86535	10
	Total	55.1250	13.72182	20
Total	Hot	62.7350	6.73352	20
	Cold	49.1300	12.03264	20
	Total	55.9325	11.83576	40

Based on the calculated averages for the primary impact, (demonstrated in the above image), the study's query becomes, “Is the distinction among 56. seventy-four and 55. a hundred twenty-five statistically extensive? in different phrases Is the primary impact of Brand statistically extensive?”



Between-Subjects Factors

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Total	Hot	62.7350	6.73352	20
	Cold	49.1300	12.03264	20
	Total	55.9325	11.83576	40

the way to perform two way ANOVA in SPSS

This means for the second one primary impact, Water Temperature.

The essential outcomes of water also are demonstrated. We also need to recognize if there's an extensive distinction between the averages of 62.735 and 49.13.

The subsequent segment in the output indicates the outcomes for Levene's Test.

Levine's test in SPSS

Levene's Test of Equality of Error Variances[a,b]

		Levene Statistic	df1	df2	Sig.
Cleanliness	Based on Mean	1.200	3	36	.324
	Based on Median	1.110	3	36	.358
	Based on Median and with adjusted df	1.110	3	29.037	.361
	Based on trimmed mean	1.184	3	36	.329

The null speculation for Levene's test is one of the same variances for all groups. Therefore, you do not need to dismiss the null speculation. All of our p-values are very large, approximately 30%, so the hypothesis of the same variance has been met. If you dismiss the null on this step, you need to not

maintain analyzing outcomes from the Two Way ANOVA in SPSS: they may be null.

Correlation Analysis

The bivariate Pearson Correlation calculates the resilience and direction of linear relationships between pairs of continuous variables.

The bivariate Pearson Correlation generates a sample correlation coefficient, r, which calculates the resilience and direction of linear relationships between pairs of continuous variables. By expansion, the Pearson Correlation analyzes whether there is statistical evidence for a linear relationship among the same pairs of variables in the population, denoted by a population correlation coefficient, ρ (“rho”). The Pearson Correlation is a parametric measure.

This measure is also called:

Pearson’s correlation

Pearson product-moment correlation (PPMC)

General Uses

The bivariate Pearson Correlation is generally used to estimate the following:

1. Correlations between sets of variables
2. Correlations within and among pairs of variables

The bivariate Pearson correlation demonstrates the following:

1. Whether a statistically substantial linear relationship occurs among two continuous variables.
2. The resilience of a linear relationship.

Data Criteria

To use Pearson correlation, your data must satisfy the following criteria:

1. Two or more consecutive variables (i.e., period or percentage level).
2. Cases must possess non-missing

values on both variables.

3. A linear relationship between the variables
4. Independent cases (i.e., independence of observations)

a. There is no relationship between the values of variables between cases. This means that:
b. the values for all variables across cases are unconnected.
c. for any given case, the value for any variable cannot affect the value of any variable for other cases.
d. no case can affect another case on any variable
e. The bivariate Pearson correlation coefficient and conforming importance test are not active when independence is infringed.

5. Bivariate normalcy.

6. Each set of variables is bivariate typically dispensed.

Each set of variables is bivariate generally allotted at all levels of the other variable(s)

This hypothesis ensures that the variables are linearly relevant; infringements of this hypothesis may imply that non-linear relationships among variables occur. Linearity can be examined visually utilizing a scatterplot of the information.

Spontaneous sample of information from the population

No outliers

The null hypothesis (H0) and alternative hypothesis (H1) of the significance test for correlation can be formulated in the following ways, being sure of whether a one-tailed or two-tailed test is needed:

Two-tailed significance test:

H_0: $\rho = 0$ ("the population correlation coefficient is 0; there is no association")
H_1: $\rho \neq 0$ ("the population correlation coefficient is not 0; a nonzero correlation could exist")

One-tailed significance test:

H_0: $\rho = 0$ ("the population correlation coefficient is 0; there is no association")
H_1: $\rho > 0$ ("the population correlation coefficient is greater than 0; a positive correlation could exist")
OR
H_1: $\rho < 0$ ("the population correlation coefficient is less than 0; a negative correlation could exist")

where ρ is the population correlation coefficien

Data Set-Up

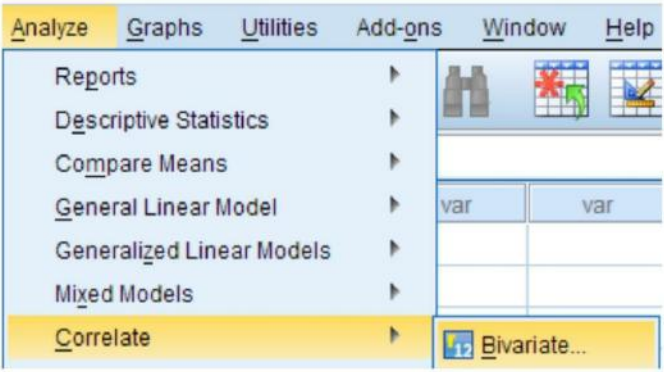
Your dataset should contain two or more consecutive numeric variables, each interpreted as scale, which will be used in the calculation.

Each row in the dataset should illustrate one different topic, individual, or unit. All computations taken on that individual or unit should be displayed in that row. If computations for one topic show on multiple rows -- for instance, if you have computations from several time points on distinct rows -- you should rewrite your data to a "wide"

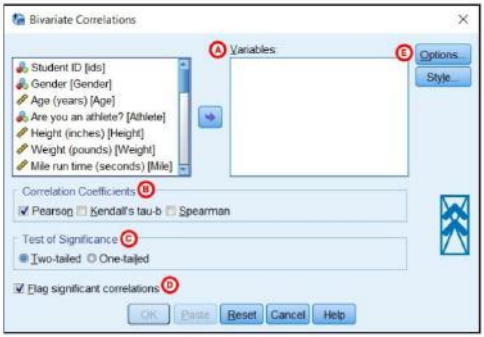
layout before you evaluate the correlations.

Perform a Bivariate Pearson Correlation

To perform a bivariate Pearson Correlation in SPSS, tap Analyze > Correlate > Bivariate.



The Bivariate Correlations window opens, where you will determine the variables to be utilized in the calculation. All variables in your dataset emerge in the record on the left side. To choose variables for the calculation, choose the variables in the record on the left and tap the blue arrow button to shift them to the right, in the Variables area.



Variables: The variables to be utilized in the bivariate Pearson Correlation. You must choose at least two consecutive variables, but may choose more than two. The test will yield correlation coefficients for each pair of variables in this record.

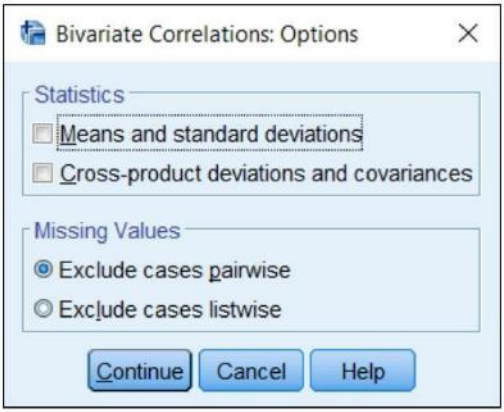
Correlation Coefficients: There are numerous types of correlation coefficients. By default, Pearson is specified. Specifying Pearson will yield the test statistics for a bivariate Pearson Correlation.

CTest of Significance: Tap Two-tailed or One-tailed, relying on your desired significance test. SPSS utilizes a two-tailed test by default.

Flag significant correlations: Checking this

alternative will contain asterisks (**) close to statistically substantial correlations in the output.

options: Tapping Options will open a window where you can determine which Statistics to incorporate and how to handle Missing Values.



Regression Analysis

Multiple regression is an extension of easy linear regression. It is used when you need to foretell the value of a variable based on the value of variables. The variable you need to foretell is known as the established variable (or sometimes, the outcome, goal, or criterion

variable). The variables we're the use to expect the fee of the established variable are known as the independent variables (or occasionally, the predictor, explanatory, or regressor variables).

Multiple regression also enables you to deduce the general fit of the version and the relative contribution of every one of the predictors to the overall variance.

Hypotheses

When you decide to examine your information using more than one regression, a part of the technique includes checking to ensure that the information you need to examine can certainly be analyzed using this method of egression. You want to do that due to the fact it's far only suitable to apply more than one regression if your information "passes" 8 hypotheses that are required for more than one regression to offer you a logical result. These hypotheses are:

1. Your dependent variable has to be assessed on a continuous scale (i.e., it is either a period or percentage variable).

2. You have more independent variables, which can be either continuous (i.e., a period or percentage variable) or categorical (i.e., an ordinal or nominal variable).

3. You have to have independence of observations (i.e., independence of residuals), which you could easily test using the Durbin-Watson statistic, which is an easy check to perform using SPSS Statistics.

4. There needs to be a linear connection between (a) the dependent variable and every independent variable, and (b) the dependent variable and the independent variables altogether.

5. Your information needs to reveal homoscedasticity, that is where the variances along the line of best fit stay comparable as you pass alongside the line.

6. Your information needs not to display multicollinearity, which happens if you have more independent variables which are fairly associated with every other variable.

7. There have to be no considerable outliers, excessive influence factors, or fairly

influential factors.

8. Finally, you want to test that the errors are generally allocated.

Example

A fitness investigator needs to be able to foretell "VO2max", a pointer of health and fitness.

Data Setup in SPSS Statistics

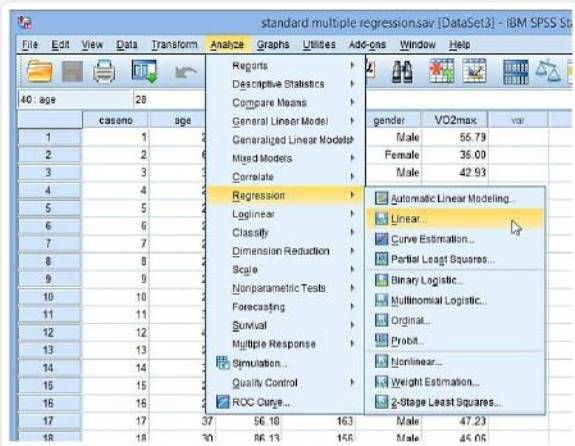
In SPSS Statistics, we developed six variables: (1) VO2max, which is the maximal aerobic ability; (2) age, which is the participant's age; (3) weight, which is the participant's weight ; (4) heart_rate, which is the participant's heart rate; (5) gender, which is the participant's gender; and (6) case no, which is the case number. The case no variable is used to make it easy for you to remove cases.

Test Procedure in

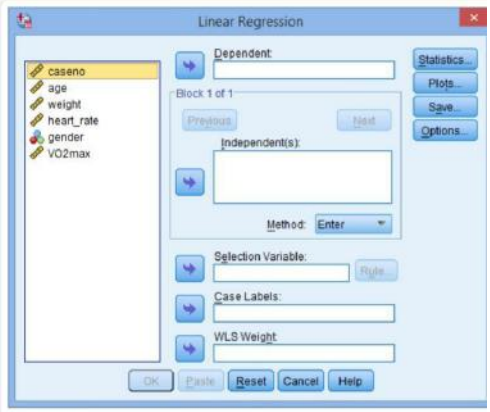
SPSS Statistics

The seven steps below reveal how to analyze your information using multiple regression in SPSS calculation when all the eight assumptions have been met.

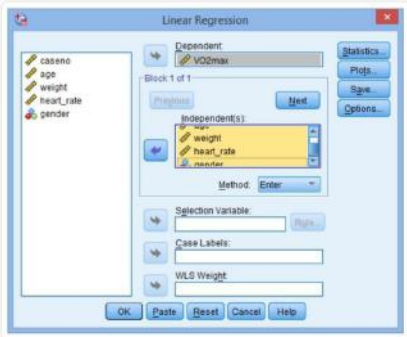
1. Tap Analyze > Regression > Linear on the major toolbar, as indicated below:



2. A Linear Regression dialogue box will be displayed as shown below:

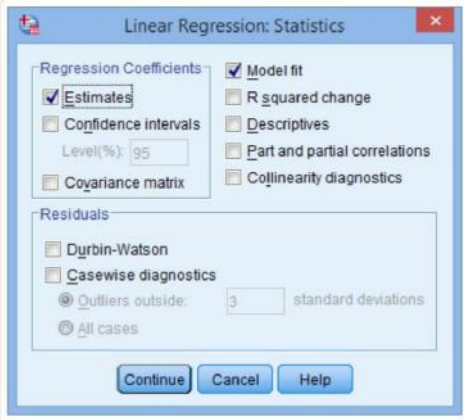


Transfer the dependent variable, VO$_2$max, into the Dependent: box and the independent variables, age, weight, heart_rate and gender into the Independent(s): box, using the buttons, as shown below (all other boxes can be ignored):



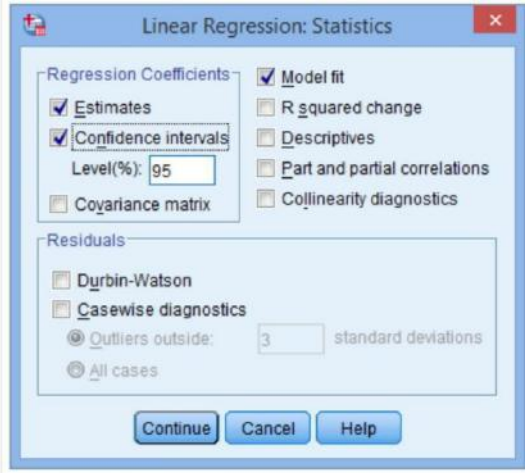
Note: For a standard multiple regression you should ignore the Previous and Next buttons as they are for sequential (hierarchical) multiple regression. The Method: option needs to be kept at the default value, which is Enter. If, for whatever reason, Enter is not selected, you need to change Method: back to Enter. The Enter method is the name given by SPSS Statistics to standard regression analysis.

3. Tap on the Statistics icon. The Linear Regression will be displayed as seen in the screenshot below.



In addition to the alternatives that are specified by default, choose Confidence intervals in the –Regression Coefficients–

region allowing the Level(%): option to remain at "95". Your screen will look like the screenshot below.



Tap the Continue icon. You will be redirected to the Linear Regression dialogue box.

Tap on the OK icon. This will create the output.

About The Author

Billy York is a data scientist with more than a decade of experience gathering and analyzing large amounts of data using the popular SPSS. He is also a lecturer in a well-known American University teaching and consulting data science. He resides in Huston, Texas with his wife Kate and three adorable kids. He attended college and graduated with a Bachelor's degree in Data Science.

Billy is a great tutor, blogger, and writer and has written so many publications on data science. When he is not writing, consulting, or tutoring, he spends time hiking, camping, and enjoying the outdoors.